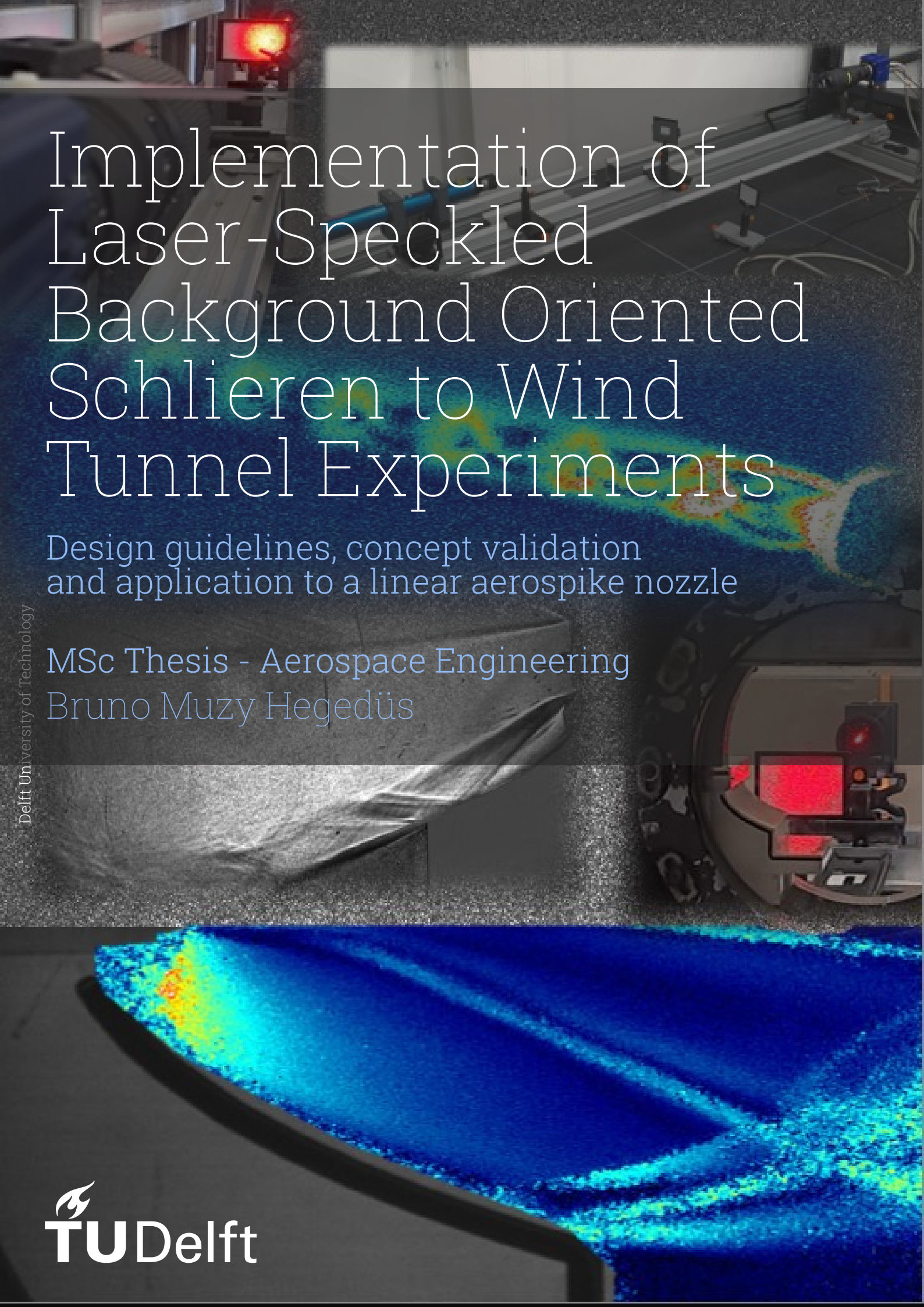




Implementation of Laser-Speckled Background Oriented Schlieren to Wind Tunnel Experiments

Design guidelines, concept validation
and application to a linear aerospike nozzle

MSc Thesis - Aerospace Engineering
Bruno Muzy Hegedüs

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Bruno Muzy Hegedüs

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Preface

A preface...

*Bruno Muzy Hegedüs
Delft, September 2026*

Summary

A summary...

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Nomenclature

Abbreviations

Abbreviation	Definition
AirBOS	Airborne Background Oriented Schlieren
BOS	Background Oriented Schlieren
BOSCO	Background Oriented Schlieren using Celestial Objects
CC	Cross-Correlation
CFD	Computational Fluid Dynamics
CoC	Circle of Confusion
DoF	Depth of Field
FCD	Fourier Demodulation
ILS	Iterative Least Squares
OF	Optical Flow
PIV	Particle Image Velocimetry
SBOS	Laser-Speckled Background Oriented Schlieren
SSTO	Single Stage To Orbit

Symbols

Symbol	Definition	Unit
a	Speed of sound	[m/s]
A	Molar refractivity	[m ³ /mol]
c	Speed of light in a medium	[m/s]
c_0	Speed of light in vacuum	[m/s]
CoC	Circle of confusion, image plane	[m]
CoC_{obj}	Circle of confusion, transferred to the object plane	[m]
d_f	Distance from the lens to the in-focus plane	[m]
d_{GG}	Distance from the beam expander to the ground glass diffuser	[m]
d_N	Nozzle exit diameter	[m]
D	Lens aperture diameter	[m]
f	Lens focal length	[m]
$f_{\#}$	Lens f-stop number	[-]
G	Gladstone-Dale coefficient	[m ³ /kg]
h	Specific enthalpy	[J/kg]
H	Height of the in-focus object	[m]
H'	Height of the out-of-focus object	[m]
IW	Interrogation window size	[px]
l	Defocus distance; positive behind, negative in front of the object	[m]
l_{px}	Pixel pitch, the physical size of one sensor pixel	[m]
$L_{FoV_{obj}}$	Field of view at the refractive object	[m]
L_{sens}	Camera sensor dimension	[m]
m	Distance from the lens to the refractive object	[m]
M	Mach number	[-]

Symbol	Definition	Unit
M_n	Mach number normal to the shock	[-]
M_{ag}	Optical magnification of the in-focus plane	[-]
M_{agobj}	Blurred object magnification	[-]
n	Refractive index	[-]
n_o	Refractive index of the ambient medium	[-]
o	Interrogation window overlap	[-]
p	Pressure	[Pa]
s_{min}	Smallest distinguishable feature size	[m]
S	Sensitivity	[m]
SR	Number of resolvable features across the object field of view	[-]
T	Thickness of the angled glass plate	[m]
u	Velocity component normal to the shock	[m/s]
V	Velocity	[m/s]
W_M	Mean molecular weight	[kg/mol]
Z	Distance from the lens to the image plane	[m]
β	Oblique shock wave angle	[°]
γ	Ratio of specific heats	[-]
δ_t	Blur circle as formulated by Gojani & Obayashi	[m]
Δs	Speckle size	[m]
Δ_t	Sampling cell in the sensor plane (Gojani & Obayashi)	[m]
Δx	Displacement measured in the image	[m]
Δy	Displacement of the background pattern	[m]
ε	Light deflection angle	[rad]
θ	Flow deflection angle; angle of the glass plate	[°]
λ	Wavelength of light	[m]
μ	Mach angle	[°]
ρ	Density	[kg/m ³]
ρ_{px}	Pixel density at the in-focus plane	[px/m]
ρ_{pxobj}	Pixel density at the refractive object	[px/m]
τ	Compressibility coefficient	[1/Pa]

Introduction

In 1935 aerodynamicists from all over the world gathered in Rome to discuss the behavior of air when objects reach high speeds, notably, the change in air density by the phenomenon of compressibility [27]. At the usual low subsonic speeds of the time, it was assumed that the variation in density was negligible. However, air density starts to considerably change with higher velocities, leading to an additional variable that needs to be accounted for and coupling the momentum and energy equations. For speeds faster than sound, these density gradients become even more relevant with the formation of shock waves, which represent an almost discontinuous change in the fluid thermodynamic properties. The developments of this conference completely changed the scope of flow physics and gave rise to a new era of aerospace vehicles; it was the dawn of compressible aerodynamics.

The study of compressible flows through experimental methods is still fundamental to the modern aerospace sector [27]. Many applications such as transonic, supersonic and hypersonic vehicles, aircraft engines and the energy sector, still heavily rely on experimental data from practical setups to aid its design and validate Computational Fluid Dynamics (CFD) models [41]. These setups require a method to visualize and quantify the variation of density in the flow. These techniques should be quantitative, non-intrusive, capture a large field of view, have sufficient spatial resolution, sufficient sensitivity and if possible, be simple. To meet these requirements, optical flow measurement techniques which rely on the behavior of light through the studied flow have been developed, such as Shadowgraphy, Schlieren, Interferometry and most recently Background Oriented Schlieren (BOS).

These optical methods designed to visualize density gradients in compressible flows are based on light's refraction. The phenomenon refers to the bending of light when it encounters differences in the medium it is traveling in, such as a fluid. These differences are quantified by the fluid's refractive index n , which is dependent on the wavelength of light and the fluid's thermochemical state. In gases, this dependency is given by the Gladstone-Dale equation, a simplification of the Lorentz-Lorentz equation, which makes explicit the proportionality between n and the gas density [43]. In nature, this phenomenon can be seen when temperature gradients lead to air density variations, creating visual distortions such as mirages and background shimmer. By controlling the medium variables in a laboratory setting, this distortion can be quantified to infer a specific property of the medium, and can be used to study, in a non-intrusive way, phenomena where gradients of the fluid thermochemical state are present.

BOS takes on the same principles of the traditional schlieren technique and improves it using modern tools. While traditional schlieren infers the density mainly using analog equipment and requiring large optical elements, BOS provides a whole field method on a simpler setup by applying cross-correlation algorithms to quantify the distortion of a reference background [46, 43]. This improvement upon traditional schlieren made BOS a widely used technique since its introduction 25 years ago, nonetheless, limitations regarding sensitivity, spatial resolution and optical configuration remain.

Laser-Speckled Background Oriented Schlieren (SBOS) emerges as an alternative capable of overcoming several of the constraints posed by BOS. This new method employs a laser-speckled pattern as the reference, which is always in focus, allowing for dynamic positioning of the background image

by adjusting the camera focus. This opens the possibility of unusual reference placements, such as in front of the test section or even inside of it, that might lead to higher sensitivity while maintaining the spatial resolution of a standard BOS configuration [32]. Moreover, the high luminous intensity of the laser generated background makes it appropriate for fast and low signal to noise phenomena, which makes it suitable for the research of reactive flows [34]. These advantages compared to standard BOS makes the study of the method interesting in an academic standpoint and potentially useful for a myriad of industrial applications.

However, the technique is still relatively unexplored compared to classical BOS systems and lacks a coherent theoretical framework for its implementation. Key aspects such as optimal setup design guidelines, uncertainty quantification, and the implications of unusual background placements have yet to be thoroughly addressed in the literature. As will be established in the theoretical review of Chapter 3, these open questions represent a meaningful gap in the current state of the art, motivating the need for a structured investigation of the method and its application in new study cases.

Following the current research in compressible flows at TU Delft, this thesis will implement SBOS in the visualization of the interaction between the freestream flow and the exhaust flow of a linear aerospike nozzle. Aerospike nozzles are able to better adapt to variations of ambient pressure during the rise in altitude of a launcher vehicle than the current bell-shaped rocket nozzle. This characteristic makes them an interesting solution to vehicles designed to reach orbit in a single stage (Single Stage To Orbit vehicles, or SSTO) as it promises an increase in efficiency of up to 15% compared to non-adaptive nozzles [17]. SBOS increased sensitivity and spatial resolution makes it promising to visualize the intricacies present in the shear layer interaction between the freestream and exhaust flow. Current literature has yet to implement SBOS in a supersonic wind tunnel environment, making this experiment the first of its type.

The research objective of this thesis is therefore to study the compressible flow in an aerospike exhaust through the implementation of the SBOS method in a supersonic wind tunnel. A foundation and practical guidelines to future research will be created by designing, building, tuning and investigating uncertainties in a simple setup as proof of concept. The results will be used to verify the methodology. Later, the knowledge and experience gathered will be used to prepare the method on a more complex test stand, TU Delft's high speed wind tunnel, to visualize the intricate interactions present at an aerospike nozzle exhaust flow. To meet this objective, the following research questions are posed:

- How can SBOS be systematically implemented to study compressible flows in a laboratory and wind tunnel setting?
 - What should be the practical guidelines for the design of a SBOS setup, balancing the trade-offs between sensitivity and spatial resolution?
 - What are the dominant sources of uncertainty in SBOS measurements?
 - How does SBOS compare in sensitivity, spatial resolution and accuracy to standard BOS?
 - How does positioning the reference inside the test section affect the results' sensitivity and data treatment?
 - What are the challenges and limitations of applying SBOS in a high-speed wind tunnel environment to visualize the flow of a linear aerospike?

The next chapters of this thesis are structured as follows: Chapter 2 provides the theoretical background; Chapter 3 provides a literature review; Chapter 4 provides practical guidelines to design a BOS setup; Chapter 5 explains the experiments conducted; Chapter 6 presents the results and data analysis; Chapter 7 discusses and validates the findings, summarizing the conclusions of the project.

2

From Compressibility to Light Refraction

A brief theoretical background is needed to understand the relationship between compressibility and refraction, to make clear how all these concepts culminate on optical flow visualization techniques. The next subsections will treat this, explaining the importance of the Mach Number, a description of shock waves and the relationship to density. Then on refraction, it will highlight the most important aspects that make Schlieren imaging possible.

2.1. Compressibility and Density

The ability of fluid to change its density is called compressibility. Air is a gas mixture, so its molecules are distanced enough that they are able to move closer together when some pressure is applied. Due to this, air is classified as having high compressibility. When objects move slowly through it, changes in pressure throughout the flow field are small enough that the effect can be neglected as an engineering approximation, and density can be assumed constant. This is the incompressible assumption, and it can be expressed through the relation between density ρ and pressure p via the compressibility coefficient τ , as shown in the equation: $d\rho = \rho\tau dp$.

When high speeds are achieved, the pressure gradients through the flow field become larger, leading to a significant change in density that needs to be accounted for. A normal threshold for compressible flows is when the Mach Number (M) surpasses 0.3, which marks the point where density changes are higher than 5% [27]. The Mach number is a non-dimensional quantity, defined by Equation 2.1, that governs the behavior of waves traveling faster than the speed of sound. This behavior was discovered in 1885 by Ernst Mach and Wentzel when working on such structures with schlieren imaging [46].

$$M = \frac{V}{a} \quad (2.1)$$

These supersonic waves were described before by Riemann, and they form when an object moves faster than the sound waves emitted by it. This makes the sound waves pile up and coalesce, forming a standing wave in front of the body [27], as illustrated in Figure 2.1.

Figure 2.1 shows an object leaving point A at a speed faster than the speed of sound, after a time interval t it has traveled Vt , while the first emitted wave has only propagated a distance at . A supersonic wave is created tangentially to the emitted sound waves and has an oblique angle μ called the Mach Angle. The picture allows us to make the following geometric relationship:

$$\sin \mu = \frac{at}{Vt} = \frac{1}{M} \quad \Leftrightarrow \quad \mu = \arcsin\left(\frac{1}{M}\right) \quad (2.2)$$

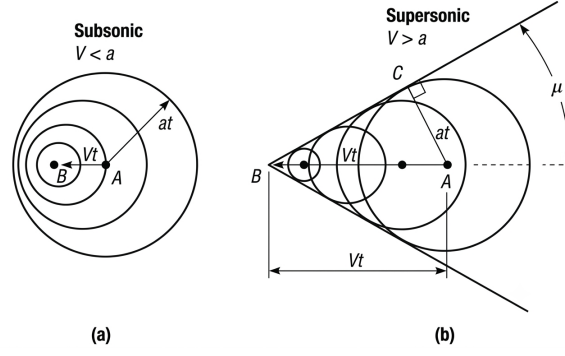


Figure 2.1: Propagation of information by an object moving at (a) subsonic speeds and (b) supersonic speeds, credit to Anderson [27]

The coalescence of the emitted waves creates an extremely thin region (of around 10^{-7}m of magnitude) where the flow properties change drastically, in an almost explosive compression process that originates a seemingly discontinuous pressure increase across it [27]. When the emitted sound from the object becomes stronger, an oblique shock wave with an angle β from the freestream air is formed, where $\beta > \mu$. Through this oblique shock wave, the pressure, density, temperature and entropy of the flow increases, while the total pressure, Mach number and velocity decreases. These relationships are described by the continuity, momentum and energy equations for oblique shocks, summarized by Equation 2.3, 2.4 and 2.5 respectively, given the geometry of Figure 2.2.

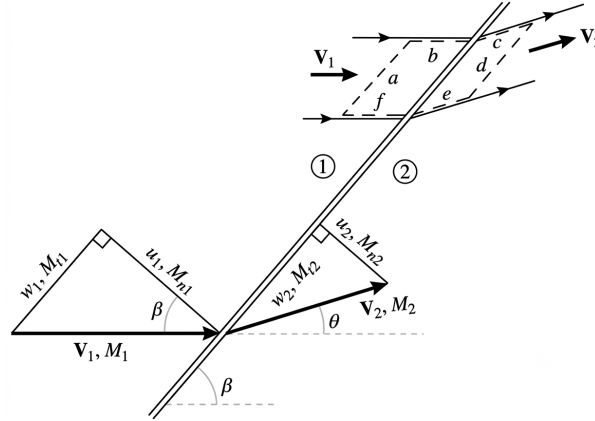


Figure 2.2: Oblique shock geometry, credit to Anderson [27]

$$\rho_1 u_1 = \rho_2 u_2 \quad (2.3)$$

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \quad (2.4)$$

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \quad (2.5)$$

These equations relate the flow properties before and after the shock. It is important to note that these changes only depend on the velocity component normal to the shock, with the tangential component maintaining constant. Considering a calorically perfect gas and adiabatic flow across the shock, it is possible to obtain the following relationships for M_2 , ρ_2 and p_2 that rely only on M_1 and β .

$$M_{n,1} = M_1 \sin \beta, \quad (2.6)$$

$$M_{n,2} = M_2 \sin(\beta - \theta) \quad (2.7)$$

$$M_{n,2}^2 = \frac{1 + M_{n,1}^2(\gamma - 1)/2}{\gamma M_{n,1}^2 - (\gamma - 1)/2}, \quad (2.8)$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_{n,1}^2}{2 + (\gamma - 1)M_{n,1}^2}, \quad (2.9)$$

$$\frac{p_2}{p_1} = 1 + \left(\frac{2\gamma}{\gamma + 1} \right) (M_{n,1}^2 - 1). \quad (2.10)$$

These are important equations, as they allow the computation of the conditions after the shock by only knowing the initial flow conditions. Another important relationship defines the flow direction θ after it has passed through the oblique shock, called the $\theta - \beta - M$ relation. This equation gives two solutions for flows over concave wedges, defining a weak shock and strong shock pairing of M_1 and β . The classification of a shock as strong or weak relates to the pressure drop across it, the bigger the total pressure drop, the stronger the shock. Consequently, the bigger the density gradient as well. This relationship is shown below, together with equation 2.9 it shows how strong density gradients naturally appear in supersonic flow across shock waves.

$$\tan \theta = 2 \cot \beta \frac{M_1^2 \sin^2 \beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2} \quad (2.11)$$

Up until now only compressive waves have been discussed, however, a supersonic flow can also expand through a Prandtl-Meyer expansion fan. When a supersonic flow finds a convex corner instead of a concave one, it turns away from itself, creating a continuous low pressure zone that accelerates it even further. The flow through an expansion fan is isentropic, so isentropic relationships can be used to compute the conditions after the fan if initial conditions are known. An important parameter in these scenarios is the Prandtl-Meyer Angle ν , which can be used to know how much the flow must bend in order to achieve a certain Mach number M_2 . Equation 2.12 and 2.13 summarizes how ν can be used to find the convex corner angle and subsequent flow direction θ .

$$\nu(M) = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \sqrt{\frac{\gamma - 1}{\gamma + 1}} (M^2 - 1) - \tan^{-1} \sqrt{M^2 - 1} \quad (2.12)$$

$$\theta = \nu(M_2) - \nu(M_1) \quad (2.13)$$

This section aims to explain the connection between compressible flow and density changes. Since variations in density produce corresponding variations in refractive index, shock and expansion structures can be detected using the deflection of light passing through the medium by employing schlieren methods. These techniques allow the visualization and quantification of the resulting density gradients, enabling experimental measurements to be compared with theoretical predictions. In the context of this thesis, the theory presented in this subsection will be paramount to interpret and judge the results stemming from the implementation of SBOS in a high-speed wind tunnel.

2.2. Compressible Jets

One important study case of compressible aerodynamics is the flow structure of a high pressure jet. Jets are produced by a high pressure flow coming out of a nozzle outlet that meets a lower pressure environment. When the pressure between the nozzle and the ambient air surpasses a certain threshold, the critical pressure ratio NPR_c , the Mach number at the minimum cross-section of the nozzle becomes 1 and the mass flow through the nozzle hits its maximum. The pressure ratio can continue to increase, however the throat Mach number and mass flow will remain the same for that given geometry, this condition is called choked flow. If the geometry of the nozzle after the throat diverges, the flow expands,

a favorable pressure gradient is established and the flow accelerates, increasing the Mach number. Depending on the pressure at the exit of the nozzle, a jet can be said: underexpanded, if $p_e > p_a$; perfectly expanded, if $p_e = p_a$ or overexpanded, if $p_e < p_a$. Figure 2.3 shows the visual difference between these different cases.

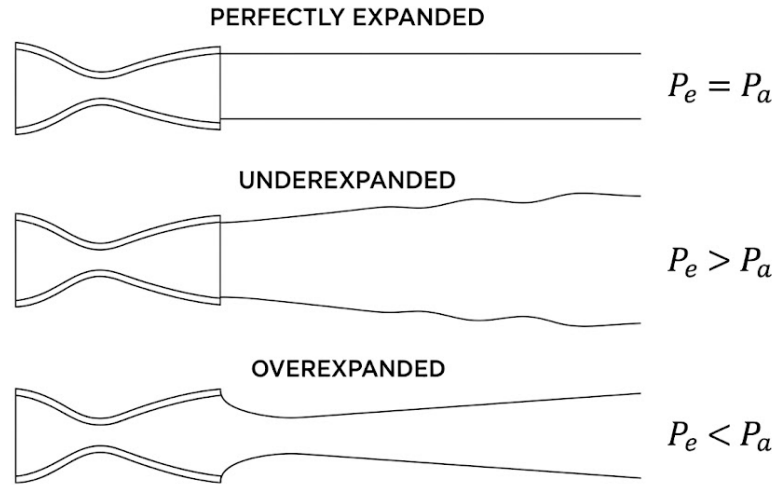


Figure 2.3: Nozzle exit flow for the three expansion conditions: perfectly expanded ($p_e = p_a$), under-expanded ($p_e > p_a$) and over-expanded ($p_e < p_a$)

Figure 2.4 shows the analytical resolution of a planar under-expanded jet. It shows that the jet is marked by periodic structures called shock cells. These cells form as a mechanism of pressure adaptation between the nozzle exit and the ambient air through alternating Prandtl-Meyer expansion and compression waves [24]. Since the pressure at the nozzle exit is greater than the ambient pressure, the flow needs to expand to match the outside pressure. The result is a Prandtl-Meyer expansion fan anchored at the lip of the nozzle. These expansion waves are reflected at the axis of symmetry into another expansion fan, which in turn reflect at the jet boundary into compression waves. These converge into one another and coalesce, forming a shock. This shock increases the pressure, leading to another round of expansion waves. This process continues until, due to friction, the jet pressure equals the atmospheric pressure.

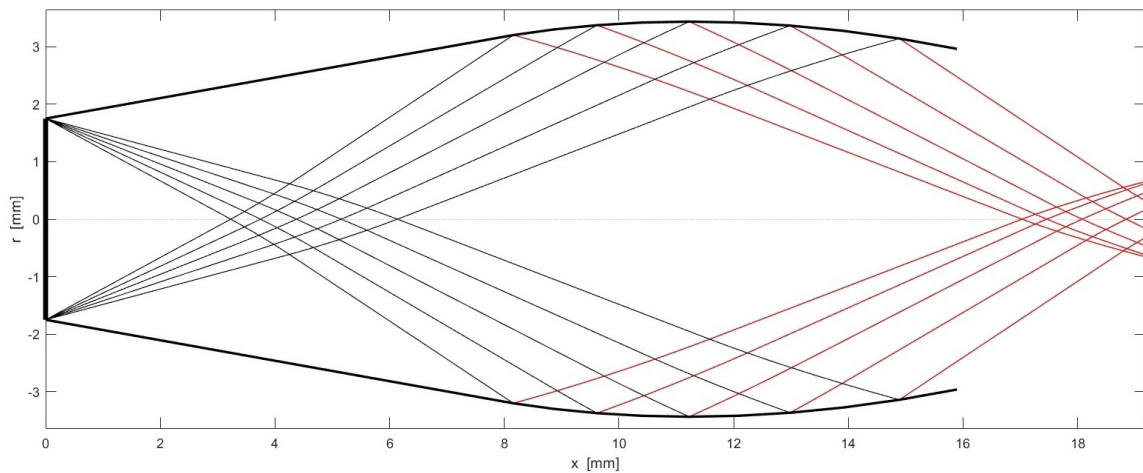


Figure 2.4: Analytical solution of a planar under-expanded jet obtained using the Method of Characteristics with $M_e = 2$ and $p_e = 2 \cdot p_\infty$. The expansion waves leaving the nozzle lips (black) cross at the axis of symmetry, reflect at the jet boundary and return as compression waves (red), which converge and coalesce into a shock in the black dots at $x = 19\text{mm}$

The three Figures 2.5a, 2.5b and 2.5c show sketches of under-expanded circular jets at different NPRs [10]. Due to its axisymmetry, the circular jet has a different structure to the planar case, displaying an

early formation of an oblique shock extending from the lip to the jet axis, often called *Intercepting shock*. When the NPR is between 2 and 4, as in Figure 2.5a, this oblique shock reflects at the jet axis and goes to the shear layer, where it turns into an expansion fan, thus forming an "X" shape at each shock cell. For higher NPRs between 4 and 7, Figure 2.5b, the intercepting shock transitions to a singular reflection and produces a normal shock called *Mach disk*, which gives each shock cell a "barrel" shape. For even higher NPRs, Figure 2.5c, the jet is dominated by a single barrel. In this case, the shock crossing the axis can no longer be considered a normal shock and its curvature must be taken into account. This last case is also marked by a decreased jet boundary diameter and a very long plume caused by entraining air.

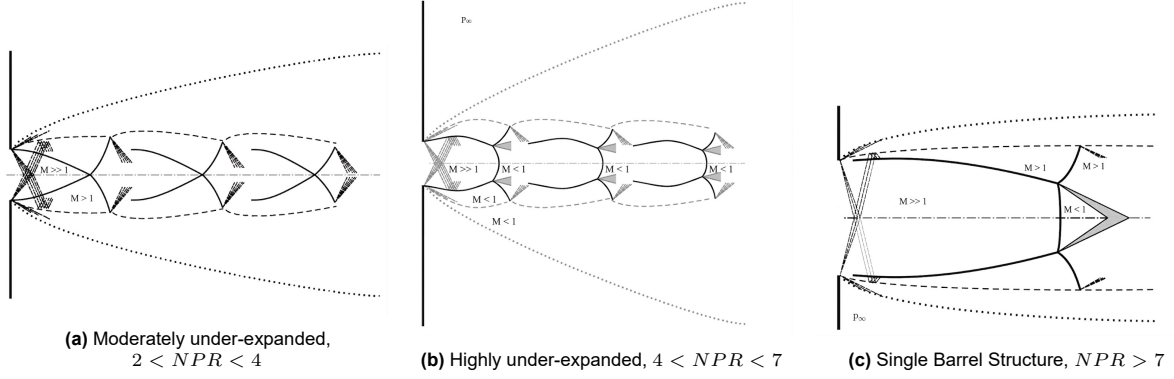


Figure 2.5: Sketches of circular under-expanded jets at increasing nozzle pressure ratio, credit to Duronio et al. [10]

As hinted by the scale of the sketches in Figure 2.5, the NPR also changes the length of the shock cells, L_s . Robert Emden was the first to correlate L_s of underexpanded choked jets with the NPR in 1899 [39, 12]. After a series of experiments he reached the format of equation 2.14, where k is a numerical coefficient that varies with the nozzle geometry and NPR range. The value of k changed from 0.77 to 1.02 during the campaign, so Emden averaged it to obtain $k = 0.88$. Later, Hartmann & Lazarus [19] introduced a constant term independent of NPR to the formulation, achieving the format of equation 2.15, with $a = 0.98$ and $b = 0.3$.

$$\frac{L_s}{d_N} = k \sqrt{NPR - NPR_c} \quad (2.14)$$

$$\frac{L_s}{d_N} = a \sqrt{NPR - NPR_c} + b \quad (2.15)$$

Since a circular jet will be used to attest the capabilities of SBOS in producing a quantitative density field, this knowledge on axisymmetric jet structure will be important to visually interpret the NPR of the results. The former sections clarify the mechanisms that lead high speed flows to vary its density. These variations influence how light bends when moving through the medium in the phenomenon of light refraction. This will be treated next.

2.3. Refraction

Refraction is the change in direction of a light ray as it propagates through a medium with spatially varying refractive index. This index is the ratio between the speed of light through vacuum, $c_0 \approx 3 \cdot 10^8$ m/s, and the speed of light through the medium, $c : n = c_0/c$.

The refractive index is a function of the light's wavelength λ , and the thermochemical state of the environment it passes through. If n is known at a certain wavelength and thermochemical state, it can be extrapolated to other states through the Lorentz-Lorentz equation, where A is the molar refractivity, dependent on the wavelength, and W_M the mean molecular weight [43].

$$\frac{n^2 - 1}{\rho(n^2 + 2)} = \frac{A}{W_M} = \text{constant} \quad (2.16)$$

The Gladstone-Dale is the linearization of this relationship for gases, where n is almost 1 [20].

$$n = 1 + G\rho \quad (2.17)$$

This relationship directly links density with the refractive index, which in turn can be given by the distortion of light through the medium. G is the Gladstone-Dale coefficient [20] [46].

As shown in Figure 2.6, when light passes from one medium to another it can behave differently depending on the refractive index field of the subsequent medium. The light's interaction between two homogeneous media, shown in Figure 2.6a, is given by Snell's Law of Refraction.

$$n_1 \sin \varepsilon_1 = n_2 \sin \varepsilon_2 \quad (2.18)$$

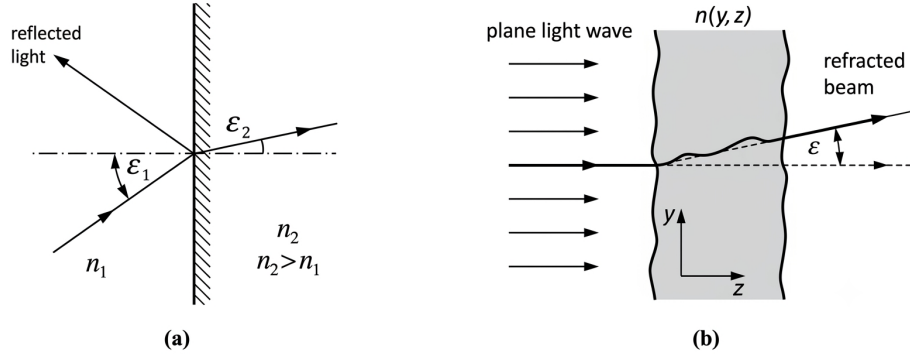


Figure 2.6: Diagrams of Light refraction: (a) Change in the beam's direction between two homogeneous media as described by Snell's Law and (b) light refraction through an irregularly distributed refractive index environment. Credit to Schmidt et al. [43]

However, for inhomogeneous environments, as in Figure 2.6b, the change in light's direction is not discrete. So a generalization must be performed to relate the deflection ε to the refractive index field. Assuming small angle deflection and small refractive index gradient, the equation below is obtained, with z being the line-of-sight direction [46].

$$d\varepsilon_x = \frac{1}{n_o} \frac{\partial n}{\partial x} dz \quad (2.19)$$

The total deflection is the integral of $d\varepsilon$ along the ray path, as is shown below. An analogous equation exists for y , by substituting it in x .

$$\varepsilon_x = \frac{1}{n_o} \int \frac{\partial n}{\partial x} dz \quad (2.20)$$

Now, if a substitution is applied with the Gladstone-Dale relationship, one can obtain:

$$\vec{\varepsilon} = \frac{G}{n_o} \int \nabla \rho dz \quad (2.21)$$

Equation 2.21 links directly the light deviation vector to the density gradient through media. Thus, in order to obtain a quantitative measure of the density field through a test section, one could employ a method to measure the displacement of light after passing through it. If the transversal refractive gradients are independent to the line of sight direction, such as a 2D planar flow, Equation 2.21 can be simplified into Equation 2.22 for x , where W is the refractive object thickness [42].

$$\varepsilon_x = \frac{G \cdot W}{n_o} \frac{\partial \rho}{\partial x} \quad (2.22)$$

The equation shows that light bends in the direction of the density gradient, that is, the direction of highest density. For a 2D planar flow, if the displacements caused by light passing through it are measured in a plane, the object density gradient can be computed and integrated to obtain a density field. For a more complex 3D flow, multiple line of-sights must be used to reconstruct a tomographic image of the density. However, some 3D flows can be reconstructed with only a 2D view due to its symmetry. An axisymmetric body is one of such case which occur frequently in various applications. The next section will demonstrate how axisymmetry can be used to obtain the 3D density field using 2D displacement vectors.

2.3.1. Axisymmetric Refractive Field

When a refractive object is axially symmetric, it is possible to reconstruct its whole density distribution with only a 2D snapshot of the light displacements through it. In such scenario, the image is a 2D projection of a 3D axisymmetric body, and the inverse Abel transform can be used to obtain its radial and axial profile. Consider the case in Figure 2.7 and Equation 2.20 for y , it can be integrated in y in both sides into Equation ??, which removes the partial derivative through the fundamental theorem of calculus.

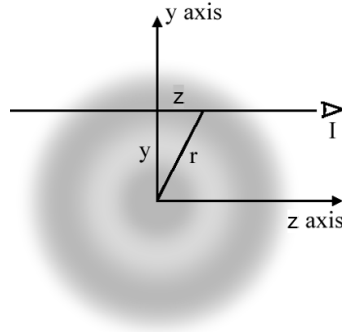


Figure 2.7: Diagram of the geometry of an axisymmetric body being seen by an observer in I. If the body, in grey, is described by a function $f(r)$, the observer will see its projection $F(y)$, which is its Abel transform

$$E_y = \frac{2}{n_0} \int_0^R [n(r) - n_0] dz \quad (2.23)$$

Now, by changing z for the radius r , E_y becomes the plain Abel transform of the refractive index n in Equation 2.24 [1]. This means that when the inverse Abel transformation is applied to E_y , a radial profile of n is obtained, which can be translated to the density through the Gladstone-Dale relation.

$$E_y = \frac{2}{n_0} \int_0^R \frac{[n(r) - n_0]}{\sqrt{r^2 - y^2}} r dr \quad (2.24)$$

This chapter has shown how changes in density can emerge in different flow conditions and how these variations modify the flow refractive properties. It is the ability to relate optical distortions to density gradients that forms the foundation of schlieren-based diagnostic techniques. The next chapter will treat these methods by presenting a brief history of how researchers have been able to measure the bending of light and translate it into qualitative and quantitative thermodynamic measures over the years, culminating in the state of the art and the topic of this thesis, SBOS.

Optical Density Measurement Techniques

Several techniques have been developed which use light refraction as a principle to visualize changes in density. These methods can be grouped into classical techniques such as Shadowgraphy, Schlieren, Interferometry and modern methods, such as Background Oriented Schlieren and its variants. This section will provide a brief historical background on classical density gradient measurement techniques and explain how these methods have developed leading up to BOS. Then, a more detailed literature review is given in BOS and SBOS to better define the state of the art of these methods and provide context to the research gap that this thesis aims to fulfill.

3.1. Classical Methods

The earliest report of a density gradient measurement technique dates back to the 1600s, when Hooke tried to visualize the flow around a candle flame with the aid of a lens [46]. His experiments resemble setups used for Shadowgraphy, where a test subject is positioned in between an observer and a light source, as shown in Figure 3.1. The variations in the refractive index at the test section bends the light coming from the source, producing shadows and brighter regions captured by the observer. Shadowgraphy is valued by its simplicity, however it provides only qualitative results, as quantitative reconstruction is difficult due to second-derivative sensitivity on the line-of-sight integration. Nevertheless, Shadowgraphy continues to be an important flow visualization technique, complementing other more complex methods in the study of compressible flows [45].

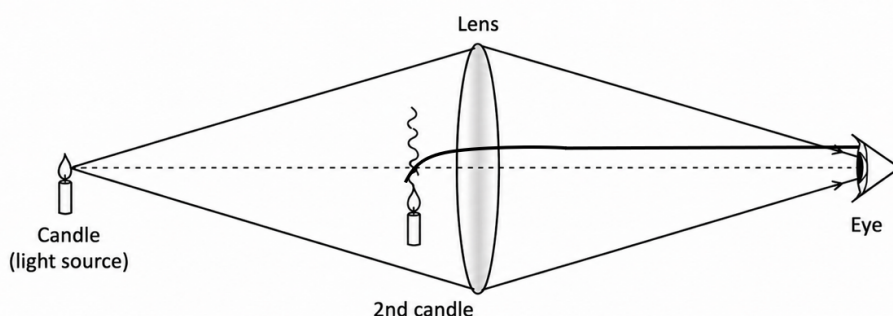


Figure 3.1: Hooke's shadowgraphy setup, credit Settles [46]

In the 19th century, Toepler coined the term "Schlieren" and was the first to apply the technique as a scientific tool [46]. Toepler's setup used a lens to focus the distorted light coming from the test section

to a point where a knife's edge was placed to block half of the image. The knife served as a filter and allowed Toepler to measure the light's deviation on the axis perpendicular to it, quantifying the density gradient in that direction. A simple Schlieren setup can be seen in Figure 3.2. The technique provides high sensitivity and has been extensively used throughout the years for the research of compressible flows, being famously implemented by Ernst Mach on the study of shock waves. However, the need for big, costly optical elements for larger scales and the sensitivity to only one axis put a constraint on the method, motivating the implementation of simpler whole-field techniques with a larger field of view. Nevertheless, the use of Schlieren is still relevant to this day and many derivations of the method exist, providing unique advantages and applications [45].

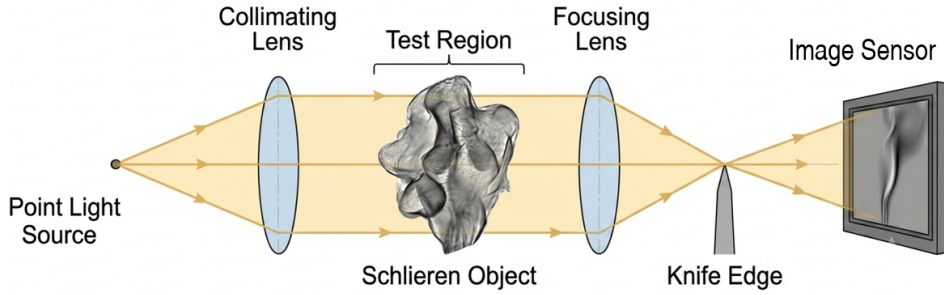


Figure 3.2: Illustration of a simple schlieren setup

Another method that relies on light deviations to measure density gradients is interferometry. In this technique, the imaging sensor is exposed to two laser beams, one passes through the test subject, which bends it and slows it down; the other one reaches the sensor unaffected. When both beams interact with the imaging sensor, they interfere, producing interference fringes that can be related to the refractive index change in the test subject. One important application of the method is the use of laser speckle photography for interferometry by Debrus [8] and Köpf [30]. They used a ground glass lens diffuser to scatter the laser beam and produce a speckle pattern, then exposing a sensitive film to this laser-speckled pattern when there is an active flow and no flow passing through it. The double exposure produces Young's fringes in the film which relates to light deviation [13]. Köpf's setup can be seen in Figure 3.3. This early use of speckle patterns foreshadows its later application in SBOS configurations.

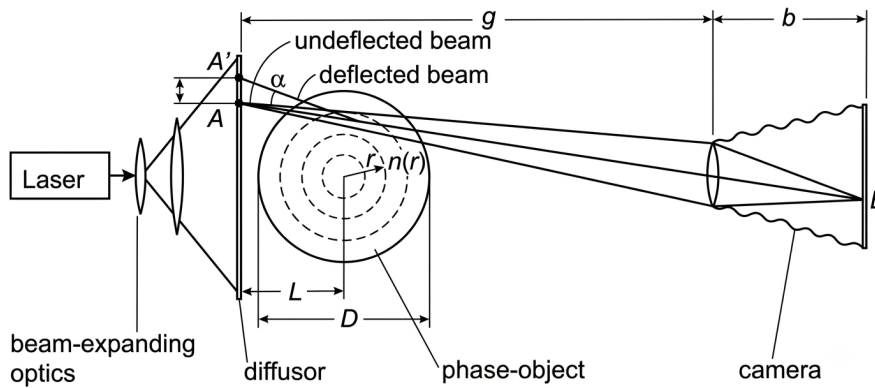


Figure 3.3: Laser-speckled Interferometry setup, credit to Köpf [30]

As aforementioned, the classical density gradient methods are still relevant to this day [45]. Shadowgraphy's simplicity gives relevant quick qualitative insights into the flow; Schlieren's high sensitivity and small post-processing requirement motivates its use in many cases, with new developments even using colored filters to get information on more than one axis at a time; and Interferometry allowing for larger fields of view than the latter two. However, each of them lack in some way, as Shadowgraphy

can only give qualitative results; Schlieren fails at larger scales and directional sensitivity; and interferometry uses a complicated setup with complex post-processing. Background Oriented Schlieren takes on the advantages of the latter methods and ameliorates its drawbacks, providing a simple, whole-field, scalable, high sensitivity technique, but still leaving some room for improvement, as will be discussed in the next section.

3.2. Background Oriented Schlieren

In the year 2000, a new technique emerged taking advantage of new sensors and image processing algorithms, the Background Oriented Schlieren (BOS) technique [41]. The method consists in measuring the distortion of a background image due to the presence of a test flow. To achieve that, a background pattern is positioned after the test section, an image of the pattern is captured when no flow is present and another when there is a density gradient that distorts the background. This displacement is quantified using image sensing algorithms and can be related to the flow's density as shown in Section 2.3 [43]. This method greatly simplifies the Schlieren setup, provides a whole field analysis of the test section and it is easier to scale than the traditional methods. These advantages, together with its accuracy and non-intrusiveness, made the technique quite popular among experimentalists, with many improved versions being proposed over the years. Figure 3.4 shows a typical BOS setup.

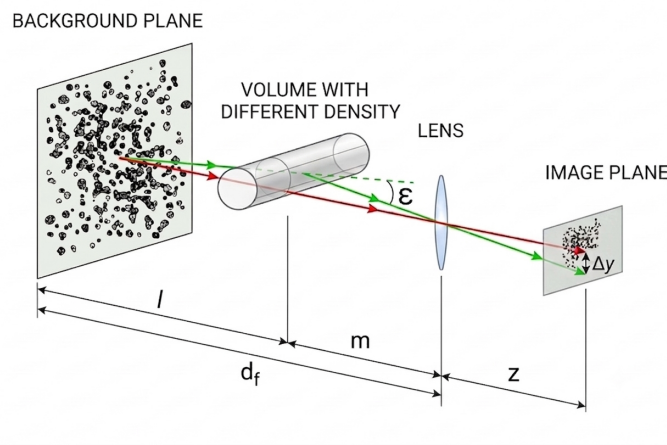


Figure 3.4: BOS setup, credit to Raffel [41]

Practical guidelines for the design of BOS setups have been established in the literature. Both Gojani et al. [14] and Schwarz et al. [44] highlighted the importance of sensitivity and spatial resolution in defining the optical configuration for BOS. Sensitivity S can be interpreted as the capacity of the setup to magnify the light distortion from the refractive object, and is set by the product of the defocus distance and the optical magnification, $S = lM_{ag}$. Spatial resolution refers to the smallest spatial scale of the studied flow that can be resolved, that is, the size of the smallest distinguishable flow feature s_{min} . It is governed by the blur of the refractive object, quantified by the circle of confusion CoC [16], its derivation can be found in Appendix A. The complete set of relations linking these two performance parameters to the physical layout of the setup is assembled in Chapter 4, where it forms the basis of the design strategy of this thesis.

Sensitivity grows with the distance between the refractive object and the background, so the first instinct is to place the background as far away as possible. The returns diminish, however: as the defocus distance grows without bound the sensitivity saturates at the focal length of the lens, $S \rightarrow f$. A longer focal length therefore raises the ceiling on achievable sensitivity, but f is itself constrained by the desired field of view $L_{FoV_{obj}}$ and the camera sensor size L_{sens} [43]. This asymptotic behavior is examined quantitatively in Chapter 4.

Furthermore, there is an inherent focusing problem on BOS. During the setup assembly, one would like to focus on the background for better contrast and position the background farther away from the test subject for increased sensitivity. But, as the distance between the test subject and the in-focus plane increases, the blur of the refractive object is worsened and hinders the spatial resolution of the

setup. One strategy would be to increase the depth of field, thus increasing the “thickness” of the in-focus plane, by increasing the $f_{\#}$ number (see Appendix A and [16]). However, reducing the lens aperture decreases the illumination reaching the camera sensor, which leads to reduced light signal and contrast. Consequently, a compromise must be made between spatial resolution and sensitivity, tuning the optical setup to achieve the maximum signal and acceptable blur.

Although it provides advantages compared to classical methods, the design of a BOS setup is marked by constraints and compromises. The achievable sensitivity is limited by the geometric configuration as increasing the sensitivity requires larger optical distances, which in turn degrade spatial resolution due to increased blur. Careful optimization of parameters is required to balance these competing effects. These limitations motivate the exploration of alternative approaches. By using Laser-Speckled Background Oriented Schlieren (SBOS) many of these shortcomings can be overcome, as it will be discussed.

Before introducing SBOS, however, further aspects of BOS deserve attention. The first corresponds to an ambiguity found in the literature regarding the optical spatial resolution. Then, it will be discussed the influence of the imaging hardware and post-processing, as both bound accuracy and resolution as much as optics do. Lastly, a range of variants that has been derived from BOS over the past twenty-five years will be presented, which gives the context for understanding both the strengths and the particular niche SBOS occupies. The next subsections will treat these matters.

3.2.1. Optical Spatial Resolution

In optical configurations, spatial resolution is limited by the amount of blur of the object. Despite the clear conceptual definition, there is no consensus on a definitive mathematical formulation of optical spatial resolution. Historically, the concept of Circle of Confusion CoC has been used to describe the blur in the image plane [16], its derivation can be found on Appendix A. Schwarz et al. [44] expanded on the concept. They experimentally tested various BOS setups and used a separability criterion to define where a visual target strip stopped being discernible from another. They then correlated that distance with the Circle of Confusion transferred to the object plane CoC_{obj} and concluded that the minimum distinguishable feature s_{min} equals half the CoC at the object plane. Meanwhile, Gojani et al. [14] derived an equation for the blur circle in the object plane by using geometry and theoretical assumptions. The two formulations do not yield the same result, so an investigation on the methods used to obtain them was conducted. Both equations are summarized in Table 3.1.

Table 3.1: Spatial Resolution Formulation by A: Schwarz et al. [44] and B: Gojani et al [14]

Author	(A)	(B)
SR formulation	$s_{min} \approx 0.5 \cdot CoC_{obj}$	$\delta_t = \frac{lM_{ag}}{f_{\#}(1+M_{ag})} + \frac{\Delta_t}{M_{ag}} \left(1 - \frac{l}{m+l}\right)$

The two formulations on the table are fundamentally different. Equation B derives the blur circle, whereas A describes a separability criterion. So if both were in agreement, A should be the radius of the blur circle B or in arithmetic terms: $\delta_t = CoC_{obj}$. In Gojani’s final formulation of δ_t , he drops one of his terms, which makes δ_t in agree with Schwarz. However, this work proves that his reasoning to drop the term containing Δ_t is weak as both terms in his formulation amount to the same value, the proof can be found in Appendix B.

To better understand the difference between the two formulations and how they relate, it is worth to explain the reasoning behind them. In his paper, Gojani makes the fundamental assumption that anything inside the CoC is indistinguishable. Therefore, the smallest distinguishable size is obtained by tracing the cone of light from the CoC in the sensor plane to a size in the out of focus object plane. Figure 3.5 shows the cone of light used by Gojani et al. in their derivation. On the image it is possible to see distances that can be translated to the convention presented in Figure 3.4, s_t corresponds to the defocus distance l , s_o is the in-focus distance d_f and s_i is the distance between the lens and the sensor Z . Equation B is already a translation from equation 12 at Gojani’s paper to this thesis’ nomenclature.

In Gojani’s derivation of the blur circle δ_t , the length in the sensor plane Δ_t is explicitly called as the “circle of confusion (CoC)” [14]. Ambiguity arises when they subsequently define it as a sampling cell $\Delta_t = k \cdot l_{px}$, where k is a small integer and l_{px} is the pixel pitch. It is not explained how k is computed

more detail. It is worth noting, that it is no use to have a higher spatial distribution of pixels if the blur limits the size of the smallest distinguishable feature. Consequently, an optimization must take place where the optical spatial resolution matches the hardware and post-processing resolution, so that the setup resources are used efficiently. Schwarz et al. make precisely this comparison by expressing both the circle of confusion and the interrogation window in the object domain and evaluating each against the size of the structure under study [44].

Once the BOS images are taken, they must be processed in a step that converts the recorded images into a displacement field data. The algorithm chosen for it bounds both the accuracy and the spatial resolution of the result. Cross-correlation (CC) was the first method applied to the technique [41] and remains the most widely used. Coming directly from its use in PIV, it divides the reference and the deflected image into interrogation windows, and uses a cross-correlation function applied to the intensity distribution of each window pair to find the displacement that best aligns them [43]. One vector is returned per window, so the spacing of the resulting field follows from the interrogation window size IW and the fractional overlap o as $IW(1 - o)$, for example, a 64 pixel window at 75% overlap yields a vector every 16 pixels.

The inheritance from PIV comes with its drawbacks, Schmidt et al. [43] point out two differences between the applications. Displacements in BOS are of the order of a few pixels rather than the ten to twenty typical of PIV [43, 40], so the dynamic range CC is designed to deliver goes largely unused while its penalty in accuracy and resolution is still present. Raffel notes that although displacements below half a pixel can still be determined with high accuracy, the smaller signal leaves BOS with relative errors of the order of 2–3% of full scale against roughly 1% for PIV [40]. Moreover, the background is manufactured in BOS as a field of discrete bright dots imitating tracer particles, which is not the best suited pattern to resolve a smooth displacement field [43, 5].

These shortcomings motivated several alternatives. Iterative least squares (ILS) uses CC to locate the integer-pixel displacement and computes the subpixel part through a separate fit. This improves subpixel accuracy and reduces sensitivity to imaging noise. Optical flow (OF) abandons windows altogether and solves for the whole field at once, returning a vector at every pixel, at a computational cost roughly an order of magnitude above CC. Fourier demodulation (FCD) recovers the deflections from phase shifts in the Fourier domain and is faster than either, but demands a periodic background such as a sinusoid or a checkerboard. On idealized synthetic images OF and FCD reproduce the ground truth almost exactly, while CC and ILS visibly undersample sharp gradients [43], a ranking supported by Cakir et al. [5], who report a clear advantage of OF over CC in both accuracy and the range of resolvable density gradient amplitudes on a Prandtl-Meyer expansion fan.

Those margins narrow considerably in practice. Schmidt et al. observe that the advantage of a higher-fidelity algorithm is realized mainly when small-scale structures dominate and the images are essentially free of noise, and that defocusing can turn the advantages obsolete by blurring the fine scales out before processing begins [43]. Schwarz et al. reach the same conclusion independently, reporting that in their own evaluations with synthetic BOS images they could not reliably match the resolution and accuracy of a 12×12 pixel cross-correlation using optical flow [44]. Since a BOS setup is deliberately defocused on the refractive object, this technicality applies directly. Combined with the periodic pattern FCD demands, which a laser speckle field cannot provide, the computational cost of OF and the absence of both from commercially available packages, the practical balance favors cross-correlation with a potential iterative least squares refinement.

Within CC, the interrogation window governs the compromise between robustness and resolution. Indeed, Schwarz et al. name the window size and the geometric blur as the two factors that jointly limit the spatial resolution of a BOS measurement [44], so the optical and processing bounds cannot be chosen independently. Small windows are preferred in BOS precisely because the displacements are small [43], yet a window holding too few features yields an unreliable correlation peak. The minimum window size recommended in the literature ranges from four to five to ten background dots, depending on the author [44]. Kähler et al. [28] quantified the resulting limit through the step response width, the smallest separation at which two vectors are genuinely independent, and showed it to grow both with the window size and the diameter of the imaged features. This means that a raise in the overlap produces a denser vector field without improving that independence, so the final window size, and not the vector spacing, is what sets the true spatial resolution of a measurement.

The same analysis determines the size of the background features themselves. Reliable subpixel interpolation of the correlation peak requires feature images larger than roughly two pixels. Below that threshold the fit can no longer locate the peak and the estimated displacements collapse toward integer values, an artifact known as peak locking [28]. Large features are equally unhelpful, since the step response width grows in proportion to their diameter, and once they reach about a quarter of the window size the partial features cut by the window edges degrade the response sharply. Holding the imaged feature between roughly two and five pixels satisfies both bounds. BOS literature converges on the same interval from experience: Raffel [40] reports background structures typically imaged at three to five pixels, with two to three preferable but rarely attainable because of diffraction at the small apertures BOS demands. He also cautions that the three-point peak fit inherited from PIV adds noise once the structures grow beyond that. Schwarz et al. quote an optimum dot image of about three pixels and worked at 3.4 to 5 pixels in their own experiments [44]. For speckle backgrounds the same target applies, Michalski et al. sized their speckles at $61.6\text{ }\mu\text{m}$ to cover roughly four pixels [33]. It is this requirement that is sought to be respected by the optical design of Chapter 4.

Processing and optics therefore impose complementary limits on what a BOS measurement can resolve. Neither has confined the technique to a single form: over the past twenty-five years it has been adapted to three-dimensional fields, to scales ranging from millimeters to aircraft in flight, and to backgrounds of very different kinds. Those developments are reviewed next.

3.2.3. Variants of BOS

Since BOS was introduced 25 years ago, the technique has not remained static, various authors have worked on modifications that enabled its application where it would be impractical to do so with the standard method. These developments were surveyed by Raffel [40] at the fifteen-year mark and, more recently, by Schmidt et al. [43] across the full twenty-five years. A significant evolution was the development of tomographic BOS, which combines multiple camera perspectives to reconstruct three-dimensional density fields [35, 40]. This method was demonstrated in applications ranging from supersonic wind tunnel tests to combustion imaging, and has since become the principal route to unsteady structures in variable-density flows [43]. For setups where multiple cameras are unavailable, plenoptic BOS offers an alternative by capturing various perspectives from a single camera through a micro-lens array, enabling independent refocusing at different planes along the line of sight from a single image [29]. These modifications have greatly expanded the scope of BOS applications beyond single 2D gradient imaging.

Other studies have focused on the limits of scaling BOS. At small scales, telecentric BOS addresses the geometric blur inherent to conventional setups through collection of parallel light rays onto the sensor [37]. This way perspective distortion is reduced and spatial resolution improved for fine-scale measurements. Van Hinsberg and Rösigen [24] developed what they called Near-Field BOS. They modified the correction factor in the BOS relation presented by Dalziel et al. [7] so that reliable densities can still be recovered when the background sits closely behind the flow. This removes the usual requirement for a large background-to-object distances. At larger scales, airborne BOS (AirBOS) has enabled schlieren imaging of supersonic aircraft in flight by using the Earth's natural surface texture as a background [48, 21], an approach whose viability rests on selecting terrain whose natural structures image at the same few-pixel scale demanded of a printed pattern [40, 43]. Another method (BOSCO) has used celestial objects such as the Sun for the same purpose [23]. Outdoor explosive and ballistic tests have also leveraged natural backgrounds and zebra board patterns to visualize shock wave propagation over fields of view inaccessible to laboratory optics [9, 43]. As a result, BOS can be applied in contexts where normal schlieren would have been impractical as it would need unreasonably heavy and large optical elements.

For high-speed applications that demand high luminous intensity, developments have been made in background illumination strategies. Back-illuminated transparent patterns and retroreflective sheeting with co-linear illumination have both been demonstrated as effective solutions in large-scale ground test facilities [22, 36]. As Raffel [40] mentions, the methods are motivated by the fact that a small aperture is wanted to limit blur but starves the sensor of light, while short exposures are wanted to limit motion blur. Therefore, retroreflective material is valuable precisely because its on-axis return does not obey the inverse-square law that penalizes a diffuse background, allowing for better contrast. The same

design pressure has driven sensor technology, from pulsed and high-framing-rate BOS [40] to event-based cameras, which record only pixel-level brightness changes, allowing them to reach kHz rates at a fraction of the illumination power and cost [43]. More recently, laser speckle has been proposed as a background source due to its high intensity and the always-in-focus nature of the speckle pattern at the sensor [32], properties that form the conceptual basis for the SBOS method investigated in this thesis.

3.3. Laser Speckled Background Oriented Schlieren

Laser-Speckled Background Oriented Schlieren is based on the same principles of BOS, with the distinct difference that a laser speckled pattern is used as reference instead of a physical background pattern. The speckle pattern is a result of the constructive and destructive interference caused by the interaction of scattered wavefronts originating from a previously collimated coherent light source, which arrive at the sensor with random phase differences. This interference pattern has been extensively investigated since the 1960s [13, 15] and has found applications in diversified fields such as the imaging of blood flow in medical fields [3], non destructive analysis of structural materials through shearography [25] and the study of density gradients using laser speckled interferometry [8]. However, the first reported use of a laser speckle pattern as a BOS background was by Meier and Rösger in 2013 [32]. It can be said then that SBOS is a relatively new development and being this recent, it is yet to be largely applied and studied, presenting the opportunity for novel and interesting developments.

Speckle creation has been extensively treated by Goodman [15] and its application to fluid mechanics measurements by Fomin [13]. As aforementioned, the speckles are created when light gets scattered by a surface rougher than its wavelength, which randomly changes the amplitude and the optical path length of the scattered rays and thus creates an interference pattern. This surface can be a rough projection plane or a diffuser such as a ground glass. The speckles can either be objective, when the scattered light propagates freely to an observation plane; or subjective, when it is collected by a lens. It is worth highlighting that speckle creation is random at its core [6, 15]. As a consequence, the theoretical mathematical treatment of this phenomenon involves complex Fourier analysis and statistics, which were considered out of scope. This thesis will treat more on the practical aspects of speckle generation and try to develop practical guidelines for its use.

The work of Meier and Rösger [32] introduced SBOS and showed that using a laser speckled background greatly expands the design space of a BOS experimental setup. By creating subjective speckles, the exit pupil of the imaging system itself can be regarded as the effective scattering spot [15]. Consequently, the speckles stay sharp at any virtual position the camera has its focus on. This permits the positioning of the pattern to be virtually anywhere along the laser beam, be it in front of the refractive object, inside of it, behind it, or even beyond the scattering plane. Consequently, optical arrangements that would be impossible in conventional BOS become feasible with SBOS. A picture of their experimental setup can be seen in Figure 3.6.

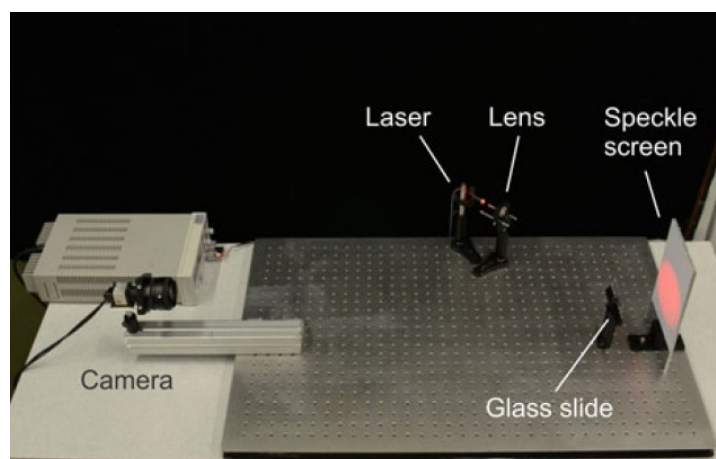


Figure 3.6: Experimental SBOS setup (single pass) used by Meier and Rösger [32]

These new optical configurations promise great benefits compared to standard BOS [32]. By focusing

the camera in front of the test section, a larger increase in sensitivity can be achieved while maintaining the same spatial resolution. Positioning the focus beyond the physical position of the screen allows for larger effective defocusing distances while maintaining the screen close to the test section, which greatly reduces the physical size of the setup. Focusing the camera inside the test section improves the spatial resolution; the implications for image quality and sensitivity, however, are not clear. Meier and Rösger also developed the double-pass mode, where the laser beam is directed coaxially with the camera through the test section, effectively doubling the sensitivity by capturing the deflection of both the illumination beam and the imaging path. The ability to choose among these optical configurations alleviates the constraints between sensitivity and spatial resolution present in standard BOS, allowing for a much more optimized optical setup.

Furthermore, replacing the conventional printed background with a laser-generated speckle pattern offers advantages in terms of background quality and spatial frequency content. Unlike physical background patterns, whose feature size is constrained by printing and manufacturing limitations, the minimum speckle diameter is only limited by the Rayleigh diffraction criterion [2]. If the interference pattern becomes diffraction limited due to the camera aperture, it will be only constrained by the laser wavelength, the f-stop number and the camera magnification. Consequently, SBOS can produce significantly finer background structures, providing higher spatial-frequency information for image correlation and potentially improving the spatial resolution.

After the work of Meier and Rösger, only a limited number of studies have explored SBOS experimentally. Bühlmann [4] employed the technique in the tomographic imaging of a heated jet near a wall where the laser-speckle was projected as a background. His measurements highlight the potential of SBOS as they would have been unfeasible with conventional BOS. Michalski et al. [33] applied the method using the in-focus plane in front of the refractive object to obtain the calorimetry of an electrical spark discharge. In contrast to the previously mentioned works that used a scattering screen, Michalski et al. used a ground glass diffuser to scatter the laser beam and pointed it directly to the camera sensor, much similar to the setup used by Debrus on speckle interferometry [8] as it can be seen in Figure 3.7. Their work also rigorously quantified the method's uncertainties and compared its accuracy with standard BOS, which will be discussed in the next section. Nakamura et al. [34] also employed laser speckles in a different way to develop a methodology intended to compute the depth location of refractive structures inside the test section and used a compressible jet to demonstrate it. Both Nakamura's and Michalski's work highlight the usefulness of SBOS for imaging fast phenomena with low signal-to-noise ratio. These studies demonstrate the method's potential, nevertheless, most of its developments have originated from a small number of research groups. Consequently, important questions regarding implementation, uncertainty, spatial resolution, and applicability in demanding environments remain insufficiently explored.

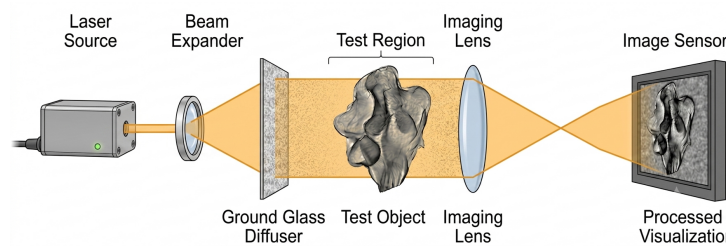


Figure 3.7: Illustration of a SBOS setup using ground-glass-diffused laser speckles, similar to Michalski's setup [33]

3.3.1. Limitations, accuracy and uncertainty of SBOS

Despite its advantages, SBOS presents some constraints and drawbacks that must be considered in its implementation. Some of these are shared with BOS such as spatial resolution and sensitivity constraints due to the chosen processing algorithms and hardware. Others are exclusive to SBOS, such as higher complexity and possible decorrelation problems. This subsection will discuss how literature treats these limitations and how it provides a comparison of the method's accuracies and uncertainties with standard BOS.

A natural complication of SBOS is the increased complexity that comes with using a laser instead of

a physical background. It requires laser compatible lenses, more optical elements, special care when handling and maintaining the setup and additional safety concerns as the laser presents a hazard at high energy settings. By implementing the method, a trade-off must be made between the increased complication and the gains in resolution and sensitivity; this will be treated in more depth in the methodology chapter.

It is also worth noting that there have been two ways in the literature to generate the speckle pattern and both have been used successfully for SBOS. It is possible to classify them into projected speckles employed by Bühlmann [4] and Meier and Rösger [32], and directed speckles used by Michalski et al. [33] and Nakamura et al. [34]. It is not clear which one of them is more effective in generating speckles, that is, what the drawbacks and advantages of each approach are. It might be only a matter of convenience when only one optical access is available, maybe one of the methods presents a more uniform distribution of speckles or is more prone to decorrelation than the other. Given the lack of literature on the topic, it would be interesting to test them both and compare them.

Decorrelation is the failure to correlate the background shift with the reference pattern and it is a phenomenon that might arise when using a laser speckled pattern as background. Bühlmann encountered this as a substantial problem on his projected laser speckle setup, which required him to use large cross-correlation windows and apply corrections for the pattern shift [4]. Li and Chiang have researched decorrelation effects for laser speckle interferometry and concluded that the main sources of the effect are surface tilt, out-of-plane displacement and lens aberration [31]. Meier and Rösger have also mentioned linear induced errors due to projecting surface tilt, so it could be that using projected speckles, like in Bühlmann's setup, makes the method more prone to decorrelation [32]. Neither Nakamura et al. nor Michalski et al. have reported the issue in their direct laser-speckle setups, which suggests that the effect could be mitigated by this approach [34, 33].

Regarding the accuracy of the technique, both Michalski and Nakamura quantified errors in their applications [33, 34]. Michalski et al. validated their implementation with a CO₂ jet and obtained a mean density for the core $0.8\% \pm 0.9\%$ lower than expected, yielding a $0.3\% \pm 0.3\%$ error for the final measured CO₂ density. Nakamura et al. described an approximated error of 10% on their method to identify the depth position of refractive objects, but did not show the errors of their density field. Compared to BOS, whose measurement error has been found to be in the order of 2–3% of the full scale by Raffel[40] and Elsinga et al. [11] reporting an accuracy of around 2% against theory for a Prandtl–Meyer expansion fan, SBOS applied by Michalski et al. performs well in terms of accuracy.

Uncertainty and noise are also important aspects to ascertain in experimental techniques. Michalski et al. [33] and Meier and Rösger [32] mention that the increased sensitivity of SBOS also makes it more prone to noise from vibration coming from setup equipment. Moreover, noise could come intrinsically from the camera sensor, room temperature gradients and even the image processing algorithm. Michalski et al. estimated the total uncertainty in their setup, quantifying partial uncertainties from the integration method, magnification, defocusing distance, cross-correlation, tomographic reconstruction and white noise, reaching a final value of 6.2%. In this project, the uncertainty sources of the designed setup will be documented and their impact quantified in Chapter 6.

3.4. Research Gap

The literature review reveals that SBOS is a promising, but not yet mature, technique. While Meier and Rösger established its conceptual foundation and demonstrated its advantages over standard BOS [32], the subsequent body of work remains sparse, originating largely from a small number of research groups and covering a narrow range of applications. This contrasts sharply with the extensive and well-validated literature surrounding conventional BOS. Three clusters of open questions emerge from this review. First, no systematic implementation guidelines exist for SBOS setup design. The sensitivity-resolution trade-off is well characterized for BOS, but whether and how those relationships translate to SBOS configurations, particularly for unusual background placements, has not been thoroughly addressed. Second, uncertainty quantification remains incomplete. Michalski et al. provide the most rigorous treatment to date [33], but their setup is specific enough that their uncertainty budget cannot be straightforwardly transferred to other configurations. Third, the behavior of SBOS in demanding high-speed wind tunnel environments is entirely unexplored, leaving open questions about

decorrelation, vibration sensitivity, and optical feasibility under those conditions.

Together, these gaps mean that a researcher wishing to implement SBOS today would find no coherent roadmap to follow. The present thesis addresses this by designing, building, and systematically investigating an SBOS setup, using a high-speed wind tunnel application as the target demonstration case. The specific research questions motivating this work are stated in Chapter 1.

4

Design Guidelines of BOS Setups

This chapter describes the methodology used to design the experimental setups. First, a brief description of the geometry of two optical setups is presented, one typical of BOS systems (Figure 4.1) and another exclusive to SBOS ones (Figure 4.2). The standard BOS setup, discussed in Section 4.1.1, places the in-focus plane on the background reference behind the refractive object, as described in the literature. Section 4.1.2 then introduces the negative defocus setup, a configuration made possible only by SBOS, where the in-focus plane is positioned in front of the test subject. For both geometries, the governing equations relate performance parameters such as sensitivity, spatial resolution and speckle size to physical setup variables.

A design strategy was then envisioned to define the setup variables and explore how they interact across the feasible design space. Rather than solving the system for a single configuration, a MATLAB script was developed to compute the full set of variables $X = (S, s_{min}, \Delta s, d_f, Z, M_{ag}, CoC, m)$ for a range of combinations of the parameters $P = (f_{\#}, l, f)$. This design space exploration, presented in Section 4.3, allows the main trends that govern the optical setup to be visualized through surface maps and provides insight into how focal length, defocus distance and f-stop number affect the trade-off between sensitivity and spatial resolution. The strategy is applied first to the SBOS configuration (Section 4.3.1), then compared to the standard BOS configuration (Section 4.3.2), before the effect of the object field of view is examined in Section 4.3.3. Finally, Section 4.5 collects the outcomes of this exploration into a single set of practical design guidelines. Together, these sections establish the reasoning behind the final design choices used in the experimental setups described in Chapter 5.

4.1. Geometries of a SBOS optical setup

Laser speckled background oriented schlieren opens the possibility of novel and unusual optical configurations that would be otherwise unfeasible for a standard BOS setup. A description of the geometry of a normal BOS configuration is given below, followed by the detailing of the negative defocus position. This section presents and compares the equations used for both setups. These formulas will be used to assess the difference in performance of a standard BOS setup and an exclusive SBOS configuration.

4.1.1. Positive defocus setup

A positive defocus setup is what is found in standard BOS configurations, where the in-focus plane is on the background reference behind the refractive object, as shown in Figure 4.1. The Figure displays relevant distances for the design of the setup: d_f corresponds to the distance of the in-focus plane to the camera lens, m is the distance between the lens and the refractive object and l is the distance between the test section and the in-focus plane, also called defocus distance. Z is the distance between the lens and the formed image, following the ideal lens relationship. The latter is important if a single converging lens is used to collect light, however it is not so relevant when compound lenses such as the ones present on modern cameras are used. Other parameters to be defined are: the magnification M_{ag} of the object in the image plane, the lens focal length f and the camera f-stop number $f_{\#}$. All these variables will be defined by the equations below.

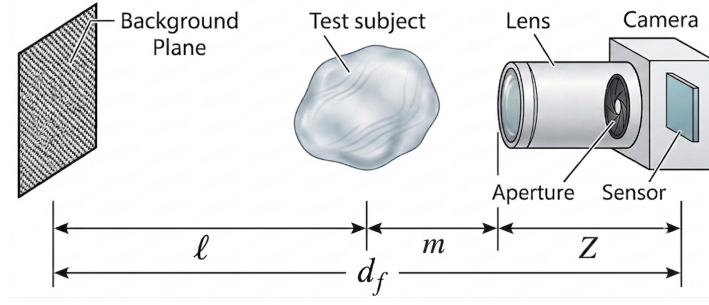


Figure 4.1: Geometry of a standard BOS Setup (positive defocus)

Equation 4.1 derives directly from Figure 4.1 and relates the distances m , d_f and l . The defocus distance l is measured from the refractive object to the in-focus plane, so it is treated as positive when the in-focus plane lies behind the object, and negative when it lies in front of it. The same equation therefore describes both geometries, with only the sign of l changing.

$$d_f = m + l \quad (4.1)$$

For a first preliminary sizing, let's consider the camera lens as ideal. This permits the use of the gaussian thin lens equation (Equation 4.2) and the coupled definition of magnification (Equation 4.3). In these equations, the object positioned at a distance d_f from the lens is in focus and has its image at a distance Z behind the lens with a magnification M_{ag} . This formula comes with a specific convention regarding the sign of M_{ag} , where negative M_{ag} corresponds to an inverted image and a positive to an upright image. Since a camera is being used, only the inverted image is going to be considered and for the sake of convenience this will be taken as positive. The camera also implies that $d_f > f$, as no image can form on the imaging sensor otherwise.

$$\frac{1}{f} = \frac{1}{d_f} + \frac{1}{Z} \quad (4.2)$$

$$M_{ag} = \frac{Z}{d_f} \quad (4.3)$$

As discussed in Chapter 3, Schwarz et al. [44] and Gojani et al. [14] have drawn guidelines for the practical design of a BOS optical setup highlighting the importance of the sensitivity and spatial resolution. Sensitivity S can be written as shown in Equation 4.4, while the minimum distinguishable feature s_{min} defined by Schwarz et al. [44] is given by Equations 4.5 and 4.6.

$$S = \frac{\Delta y}{\varepsilon} = l \cdot M_{ag} \quad (4.4)$$

$$CoC = \frac{lfM_{ag}}{f_{\#}m} \quad (4.5)$$

$$s_{min} = \frac{CoC}{2 \cdot M_{ag_{obj}}} \quad (4.6)$$

In Equation 4.6, $M_{ag_{obj}}$ is the blurred object magnification, which is defined by Equation 4.7.

$$M_{ag_{obj}} = \frac{L_{sens}}{L_{FoV_{obj}}} \quad (4.7)$$

Equation 4.7 relates the field of view of the object and the camera sensor dimension. In the literature [43, 44], many authors define the focal length f by choosing a desired field of view of the object, which is derived through the ideal lens equation and the definition of magnification. Thus it makes the fundamental assumption that the object is in focus as the ideal lens equation only applies to in-focus objects. Since the refractive object is out of focus, the assumption is fundamentally wrong and another equation must be derived to connect the lens' focal length to the blurred object's dimension. Geometry can be used to derive Equation 4.8 without relying on the ideal lens relationship, the derivation can be found on Appendix B.

$$L_{FoV_{obj}} = \frac{m \cdot L_{sens}}{f(M_{ag} + 1)} \quad (4.8)$$

4.1.2. Negative defocus setup

Figure 4.2 describes the geometry of a SBOS setting with negative defocus distance, which means that the in-focus plane is in front of the test subject. All the equations from the previous section remain valid in this case, that is, Equations 4.1, 4.2, 4.3, 4.4, 4.5, 4.6 and 4.8. The single difference is that l is now negative, so that Equation 4.1 places the in-focus plane in front of the refractive object, as shown in Figure 4.2.

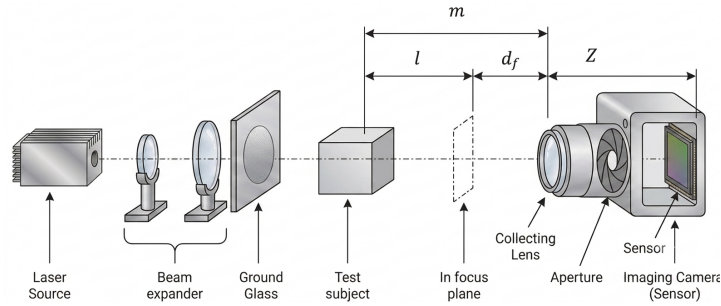


Figure 4.2: Geometry of a SBOS setup with in-focus plane before the test subject (negative defocus)

Another difference from BOS to SBOS is the speckle generation. Barker and Fourny [2] have used the Rayleigh limit to quantify the minimum speckle size when working with laser interferometry, the equation is presented as Equation 4.9.

$$\Delta s = 1.22\lambda f_{\#}(1 + M_{ag}) \quad (4.9)$$

It has been mentioned in Chapter 3 that this configuration provides a better sensitivity for the same spatial resolution than standard BOS, this might be explained through a careful analysis of the setup geometry and the equations that govern it. In Equation 4.4, S is directly proportional to M_{ag} , while in Equation 4.3, M_{ag} is inversely proportional to d_f . Comparing Figure 4.1 and Figure 4.2, it is clear that for the same defocus distance, d_f is smaller in the latter case. Consequently, M_{ag} is larger and the sensitivity is higher. However, since CoC also depends on M_{ag} , it is not immediately obvious whether the gain in sensitivity is offset by the additional blur.

4.2. Sensitivity-Resolution Ratio

The dilemma of increasing sensitivity through magnification but also increasing blur can be solved by relating sensitivity and spatial resolution directly. Using the definition of s_{min} in Equation 4.6 and substituting the circle of confusion (Equation 4.5) yields Equation 4.10.

$$s_{min} = \frac{lfM_{ag}}{2f_{\#} m M_{ag_{obj}}} \quad (4.10)$$

The field of view relation of Equation 4.8 can be rewritten as $mM_{ag_{obj}} = f(1 + M_{ag})$, which leaves Equation 4.11. Recognizing lM_{ag} as the sensitivity (Equation 4.4), the ratio between the two performance

parameters reduces to Equation 4.12.

$$s_{min} = \frac{lM_{ag}}{2f_{\#}(1 + M_{ag})} = \frac{S}{2f_{\#}(1 + M_{ag})} \quad (4.11)$$

$$\frac{S}{s_{min}} = 2f_{\#}(1 + M_{ag}) \quad (4.12)$$

Equation 4.12 shows how the sensitivity-resolution trade-off is locked by the magnification and f-stop number. Since the negative defocus configuration allows for higher magnification, a larger ratio is achieved. This means that the sensitivity will grow larger than spatial resolution and a better trade-off is reached. As an example, for $f = 0.1$ m, $|l| = 0.1$ m, $f_{\#} = 16$ and $L_{FoV_{obj}} = 2$ cm, the positive defocus solution gives $M_{ag} = 0.47$ and $S/s_{min} = 47$, while the negative defocus solution gives $M_{ag} = 1.06$ and $S/s_{min} = 66$: the sensitivity rises by a factor of 2.2 and s_{min} by a factor of 1.6, a net gain of roughly 40% in the trade-off.

Another consequence is that the f-stop offers a convenient means of improving the sensitivity-resolution trade-off. Contrary to the magnification that, given a fixed field of view, is modified by the defocus distance and camera focal length; the f-stop is easily modifiable by reducing the lens aperture diameter without the need to change any other parameter in the optical setup. This comes with two caveats, as an increase in the f-stop decreases the amount of illumination in the camera sensor, reducing contrast; and it increases the minimum speckle size through Equation 4.9.

Moreover, it is worth noting that Equation 4.12 is bound by the same variables as Equation 4.9 once the laser is selected. Since Δ_s must be kept between 2 and 5 pixels for optimal cross-correlation [28], the sensitivity-resolution ratio is confined to a fixed achievable interval. This is not a hard constraint, however, as the speckle size could be bigger than 5 pixels if the CC interrogation window is large enough, with the downside of decreased spatial resolution.

The relations derived in this subsection establish how the performance parameters are coupled, but not where in the design space a suitable configuration lies. The next section first shows how the system could be solved for a given specification, then explores the design space numerically. Surface maps are used to show how sensitivity, spatial resolution, speckle size and the setup distances respond to f , l and $f_{\#}$, and how the trends seen there are the graphical counterpart of Equation 4.12.

4.3. Design Strategy

Through the equations derived in the previous sections, a system can be solved to define the geometric aspects of a BOS and SBOS optical arrangement. The sensitivity S and the minimum distinguishable feature size s_{min} are chosen from the requirements of the application, L_{sens} is restricted by the camera and $L_{FoV_{obj}}$ is set by the object of study. Since the f-stop is a discrete variable depending on the camera lens hardware that is mostly independent of other variables, it is also specified as an input. This way, equations, 4.1, 4.2, 4.3, 4.4, 4.8 and 4.12 form a system of six equations and six unknowns M_{ag} , l , d_f , m , f and Z , which admits a unique solution. The circle of confusion and the speckle size then follow from Equations 4.6 and 4.9. The latter must be verified at the end to ensure the speckle size is in the range of 2-5 pixel, to achieve optimal cross-correlation.

It is also interesting to follow the opposite direction of the solution and assess how the performance parameters change for different optical arrangements. For that, the inputs were changed from sensitivity and spatial resolution to variables that are easier to define and modify when building the experimental setup such as the lens' focal length f and the defocus distance l . A MATLAB script was written that would solve the system for a given combinations of $P = (f_{\#}, l, f)$ and compute $X = (S, s_{min}, \Delta_s, d_f, Z, M_{ag}, m, CoC)$. A design space exploration was conducted where the parameters P were varied within feasible ranges and the system solved for each combination. The data collected allowed the construction of surface maps that permitted the visualization of the main trends permeating the design space. With this script at hand, it is possible to select the design point that gives the sensitivity, field of view and spatial resolution desired for a given application.

4.3.1. Surface Maps for negative defocus setup

Surface Maps for each variable in X were obtained to better understand the system at hand. This section will show and discuss the main trends in sensitivity, spatial resolution, M_{ag} , m , d_f and speckle size Δ_s , aiming to obtain a better preliminary insight on how to build an optimal negative defocus SBOS setup. For the sake of convenience, the maps presented here only deal with the absolute value of l . Moreover, Z and CoC have been omitted because the concept of s_{min} encapsulates CoC , and Z is not relevant for a setup with compound camera lenses. The pixel density and L_{FoV} for the in-focus plane have also been computed and will be helpful for the validation of the design strategy during the first experimental campaign, information on them can be found in Appendix ???. For this first iteration, $L_{FoV_{obj}}$ was set to 2cm, $\lambda = 632.8\text{nm}$, pixel size equal $6.5\text{ }\mu\text{m}$ and the camera sensor has 2560×2160 pixels, resulting in $L_{sens} = 1.4\text{ cm}$. Figure 4.3 shows surface maps of Sensitivity and s_{min} given a range of P . The colored isolines are illustrative reference contours and do not correspond to a selected design point: the red isoline marks $s_{min} = 0.0004\text{ m}$ and the white one $S = 0.02$.

don't forget this

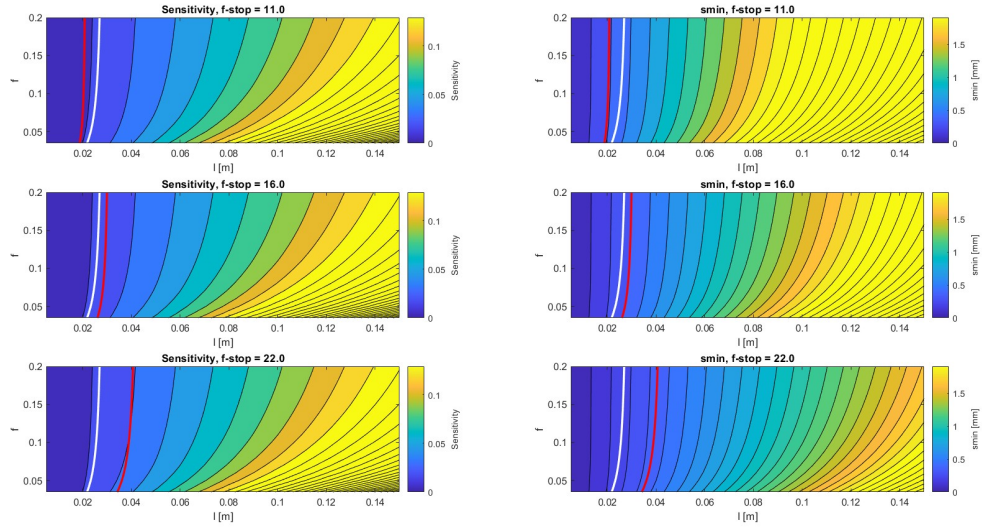


Figure 4.3: Surface maps of S and s_{min} for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$ and $l \in (0.005, 0.15)$ for a negative defocus setup, with the red isoline representing the points where $s_{min} = 0.0004\text{ m}$ and the white line showing $S = 0.02$.

From Figure 4.3, it is possible to draw the main trends in the two performance parameters, Sensitivity and s_{min} , with changing focal length f and defocus distance l . The sensitivity seems to improve with a higher l , whereas the spatial resolution is worsened, signaled by an increase in the smallest distinguishable feature size s_{min} . For a fixed l , both S and s_{min} increase with lower f but the effect becomes less prominent for higher focal length values. This means that a shorter focal length can reach the same sensitivity as a longer one at a smaller l , at the cost of a proportionally larger s_{min} . Moreover, the larger isoline gaps present at higher f means that the system performance is less prone to changes for small deviations in l , which makes it more robust to uncertainties and errors induced during the setup assembly. This means that a higher f and a lower l are desirable for an optimal optical setup.

The figure also shows that the f-stop number can be used to easily increase the setup performance, as previously mentioned. When comparing the trends in sensitivity and s_{min} for the 3 different $f_{\#}$, the red isoline moves more to the right than the white isoline when the f-stop increases. From this, it follows that a higher f-stop allows for better spatial resolution while still maintaining high sensitivity. From Equations 4.1, 4.2, 4.3 and 4.8, it can be seen that M_{ag} , m and d_f are independent of $f_{\#}$. This means that a change in $f_{\#}$ can be made by only changing it in the camera lens while maintaining the same optical setup. Consequently, the f-stop should ideally be set to the highest possible to optimize the sensitivity-resolution trade-off.

Other important parameters to analyze are the optical magnification M_{ag} and the distances m and d_f , defined in Figure 4.2. The magnification describes how amplified the image is related to the real object and, as previously determined, it influences the sensitivity-resolution. The distances define how close

the test section and in-focus plane are from the camera and directly influence M_{ag} , which makes them important practical parameters that define the setup size. Figure 4.4 presents the surface maps of these quantities for a given range of P .

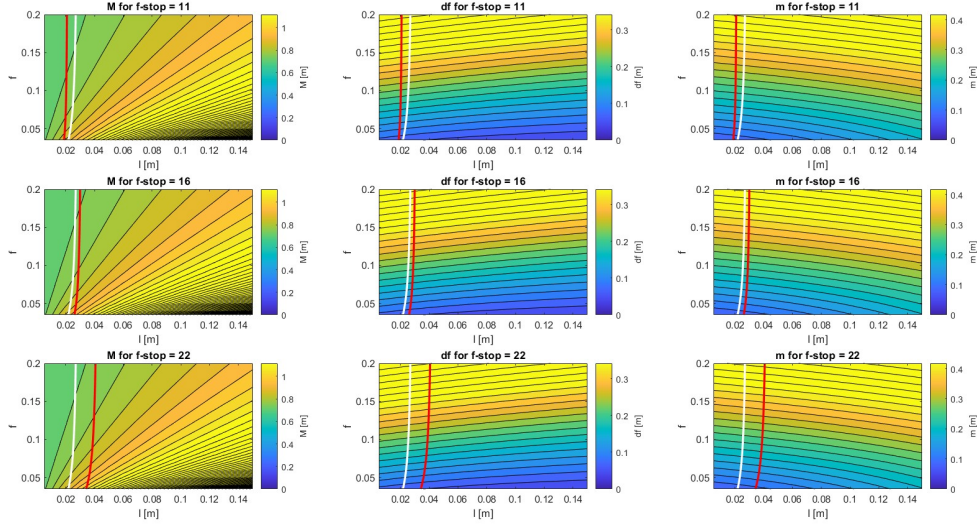


Figure 4.4: Surface maps of M_{ag} , m and df for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$, $l \in (0.005, 0.15)$ and $L_{FoV_{obj}} = 2\text{cm}$ of a negative defocus setup, with the red isoline representing the points where $s_{min} = 0.0004\text{ m}$ and the white line showing $S = 0.02$.

From Figure 4.4, the distance m only increases slightly along l but presents a steep gradient in the f axis. Together with the findings of Figure 4.3, where S and s_{min} don't vary much with f , this suggests that a smaller focal length lens could be used to obtain a more compact setup with little cost in sensitivity and spatial resolution. Recall that the gradients of S and s_{min} in relation to f become more intense with lower f . Consequently, a moderate focal length such as $f = 0.1\text{ m}$ is preferable to a considerably shorter one such as $f = 0.05\text{ m}$, where the gradients of S and s_{min} are steepest. It can also be argued in favor of $f = 0.1\text{ m}$ over a longer lens such as $f = 0.2\text{ m}$, given the savings in space. Nonetheless, the recommendations made for f earlier in this section still hold, meaning that if the dimension of the setup is not a constraint, a higher f is more desirable.

The Figure also shows the trends on optical magnification. The surface maps show that M_{ag} increases with higher defocus distance, whereas it tends to decrease with higher f . The optical magnification can be accurately measured via a calibration plate in the in-focus plane, therefore it will be an important parameter to be validated during the experiment. Moreover, Equation 4.12 shows that the sensitivity-resolution ratio grows monotonically with M_{ag} .

As discussed in Chapter 3, the minimum speckle size is also an important parameter as it limits the spatial resolution by restricting the allowable interrogation window size on the cross-correlation software. Figure 4.5 shows how the speckle size changes for a given range of P . Since Δ_s is directly dependent on the magnification through Equation 4.9, it has the same trends as M_{ag} . However, the f-stop number has a much bigger effect in Δ_s . It is recommended in the literature to limit the speckle size to 2-5 pixels to ensure optimal cross-correlation and avoid peak locking errors [28], which essentially also limits the usable f-stop numbers of the setup. In this studied case, this means that $f_{\#}$ shouldn't be bigger than 22 nor smaller than 11.

It must be said that the speckle generation process is completely random [6]. The minimum speckle size is defined by the Rayleigh criterion but the average dimension might be bigger than the expected. If it happens that the average speckle size is bigger than 5 pixels, the downside will be the need to use larger interrogation windows, therefore reducing the spatial resolution. Nevertheless, if a high magnification is being used, a big interrogation window might still lead to a small vector pitch and an acceptable resolution. Therefore, there is more to the spatial resolution than the amount of blur in the

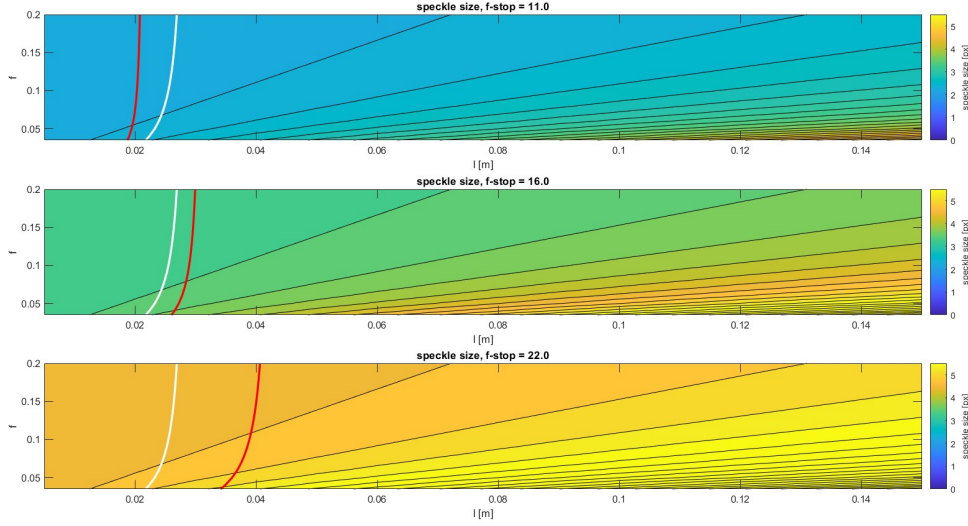


Figure 4.5: Surface maps of Δs for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$, $l \in (0.005, 0.15)$ and $L_{FoV_{obj}} = 2\text{cm}$ for a negative defocus setup, with the red isoline representing the points where $s_{min} = 0.0004\text{ m}$ and the white line showing $S = 0.02$.

image as it can also be affected by the CC post-processing. This amounts to the question: how is it possible to reconcile both resolutions such that no resource is being wasted to get a better optical resolution and obtaining a worse processing resolution? Section 4.4 answers this question and provides a design criterion to choose the best BOS setup. Before getting into the SR criterion, it is worth first analysing how negative defocus compares with positive defocus in the next section.

4.3.2. Surface Maps for positive defocus setup

Surface maps for a standard BOS setup were also obtained. These maps allow for a comparison of the trends permeating the design space of a positive defocus to the negative defocus present in the last section. Similarly to Section 4.3.1, this subsection will show and discuss the main trends in sensitivity, spatial resolution, M_{ag} , m , d_f and speckle size Δs , given $L_{FoV_{obj}} = 2\text{cm}$, $\lambda = 632.8\text{nm}$, pixel size equal $6.5\text{ }\mu\text{m}$, sensor dimension of 2560×2160 pixels, resulting in $L_{sens} = 1.4\text{cm}$. Figure 4.6 contains surface maps of Sensitivity and s_{min} given a range of P .

Through Figure 4.6, the main trends on sensitivity and spatial resolution for a positive defocus setup can be analyzed. Both parameters seem to reach an asymptote for higher l . The higher the f , the higher the asymptote value. Moreover, it seems that the sensitivity is unaffected by $f_{\#}$, while the s_{min} asymptotes moves up and right for an increasing f-stop, meaning that a higher f-stop allows for a better spatial resolution for the same sensitivity, as demonstrated by Equation 4.12. Recall the discussion from Section 3.2, summarized in Equation 4.13. The limit shows that the sensitivity asymptotes shown in Figure 4.6 will make S converge to f at large l . Consequently, no matter how far away the background is positioned in a positive defocus configuration, the sensitivity will ultimately be limited by the focal length of the camera, whereas with the negative defocus, sensitivity is unbounded. Nevertheless, while designing the setup, the same recommendation reached for SBOS also applies for BOS, a higher focal length and f-stop is the most desirable.

$$\lim_{l \rightarrow \infty} S = f \quad (4.13)$$

Regarding M_{ag} , m and d_f , Figure 4.7 shows that the positive defocus setup variables have different trends compared to a negative one. Variables m and d_f seem to have switched behaviors if compared to their trends in Section 4.3.1, with m increasing slightly with l , while d_f decreases little with l , and both increasing considerably with f . Due to the large gradient of both distances in the f axis and the nature of the S and s_{min} , similarly to the negative defocus case, it is possible to obtain a more compact setup

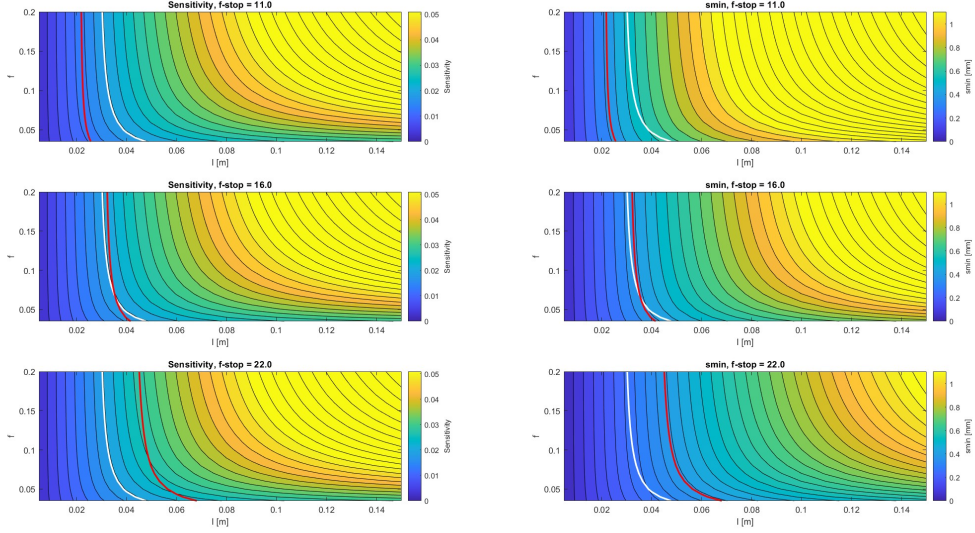


Figure 4.6: Surface maps of S and s_{min} for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$, $l \in (0.005, 0.15)$ and $L_{FoV_{obj}} = 2\text{cm}$ of a positive defocus setup, with the red isoline representing the points where $s_{min} = 0.0004\text{ m}$ and the white line showing $S = 0.02$.

using a smaller focal length if needed, with little loss in sensitivity and spatial resolution. Comparing M_{ag} from positive l to negative l , it is worth noting that the absolute values are smaller and the trend is inverted, which strengthens the argument that SBOS presents a higher S for the same s_{min} due to an increase in M_{ag} . The decrease in M_{ag} also reflects in a smaller speckle size for the BOS case through Equation 4.9, this can be seen in Appendix ??.

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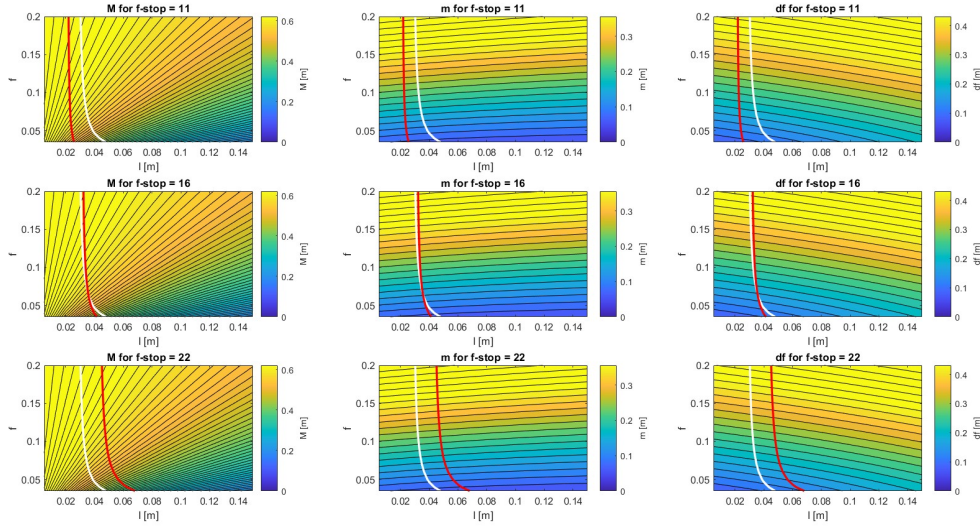


Figure 4.7: Surface maps of M_{ag} , m and d_f for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$, $l \in (0.005, 0.15)$ and $L_{FoV_{obj}} = 2\text{cm}$ of a positive defocus setup, with the red isoline representing the points where $s_{min} = 0.0004\text{ m}$ and the white line showing $S = 0.02$.

4.3.3. Effect of FoV

Even though the proof of concept experiment was done with very a high magnification on a small field of view, to generalize the presented guidelines, we'll investigate the effect of scaling field of view. Surface maps will be used to compare the trends in sensitivity and spatial resolution of negative and positive defocus setups when the $L_{FoV_{obj}}$ is changed from 2cm to 7cm. Figure 4.8 shows these graphs, surface

maps for other parameters can be found in Appendix ???. For this case, the red isoline value has been slightly modified to $s_{min} = 0.00035$ m. However it still allows the comparison with previous images given the very small difference between them.

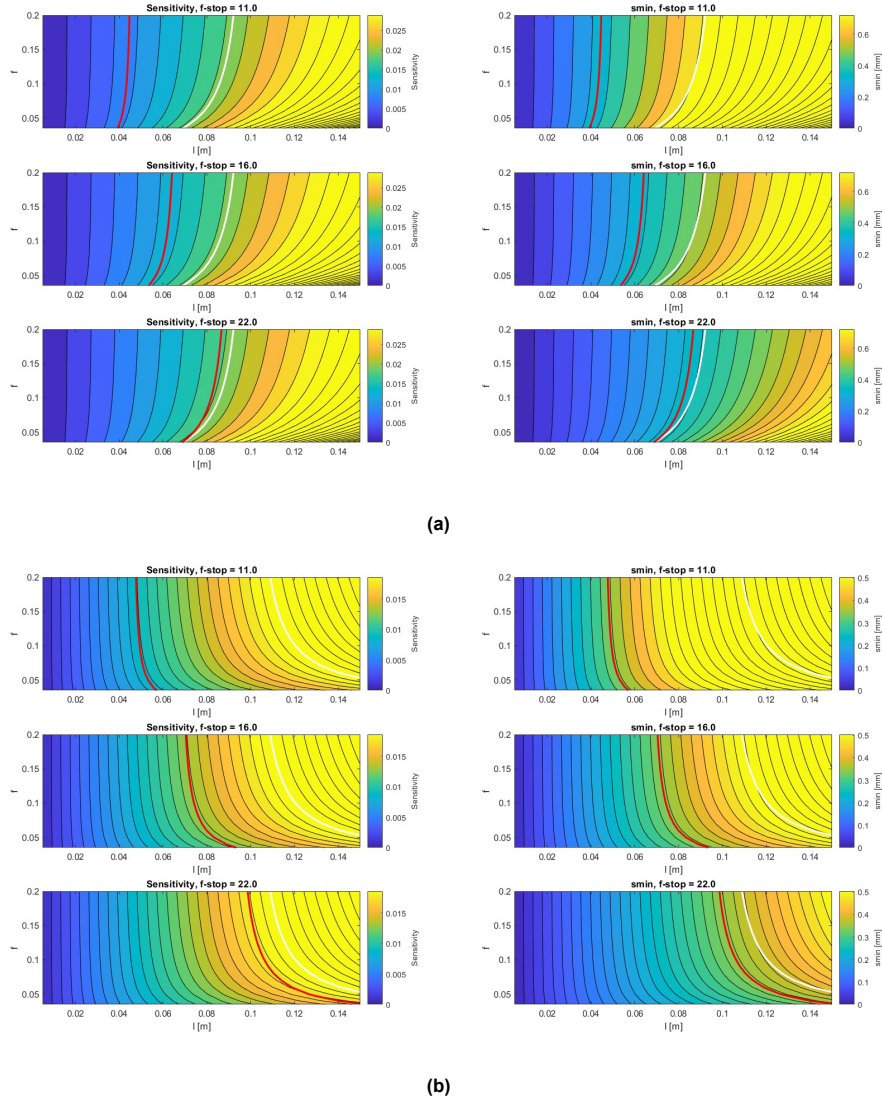


Figure 4.8: Surface maps of S and s_{min} for $f_{\#} \in [11, 16, 22]$, $f \in (0.035, 0.200)$, $l \in (0.005, 0.15)$ and $L_{FoVobj} = 7\text{cm}$ of a negative defocus setup (a) and a positive defocus setup (b), with the red isoline representing the points where $s_{min} = 0.00035$ m and the white line showing $S = 0.02$.

From Figure 4.8 some differences from the surface maps presented at Sections 4.3.1 and 4.3.2 can be highlighted. Generally, a higher field of view results in smaller S and s_{min} due to the smaller optical magnifications needed to achieve the desirable window size, as per Equation 4.3 and 4.12. This results in a much higher defocus distance needed to achieve the same S . It also translates in less susceptibility to obtain blurred images, which is signaled by the generally smaller s_{min} . This is evident from the position of the red and white isolines, which are both located at much higher l 's than their smaller L_{FoVobj} counterparts. This characteristic makes the advantage of SBOS even more prominent: with negative defocus a higher sensitivity can be achieved at lower l 's while maintaining a low s_{min} .

Up until now, the surface maps have been useful to visualize the equations displayed in Section 4.1, 4.2 and understand better the trends that exist between variables. However, a definitive criterion to choose the best optical setup is still missing. This will be done by solving a problem mentioned in Section 4.3.2, to match the optical spatial resolution and software spatial resolution so the experimenter can more

efficiently use its resources. The solution is presented in the next section.

4.4. SR criterion

To reconcile the optical spatial resolution and the post-processing resolution, it must be expressed in terms of pixels. Historically, M_{ag} has been used by researchers to provide an insight on how much detail the camera can capture, as it defines the pixel density through Equation 4.14 [43, 14]. However, since the object is blurred, it is not necessarily true that a higher pixel density will lead to a better spatial resolution as the number of pixels per millimeter may increase together with the size of the blur, as per Equation 4.5.

Moreover, given that the object is out of focus, its size differs from the one it would have if it were in focus (see Appendix B). Therefore, there are two magnifications at play in the setup, one for the blurred object and another for the in-focus plane. The blurred object magnification is fixed by the desired field of view and camera, defined by Equation 4.7. Whereas the optical magnification shown in Figure 4.4 is defined by Equation 4.3. The amount of detail truly captured by the camera is constrained by $M_{ag_{obj}}$, fixed by the defined object field of view. It also defines how many pixels can image a mm of the object, through the pixel density (Equation 4.14) and influences the blur of the image, via Equation 4.6. Consequently, a trade-off must be done, where the level of blur is balanced with the pixel density such that the setup resources can be used efficiently.

$$\rho_{px} = \frac{M_{ag}}{l_{px}}, \quad \rho_{px_{obj}} = \frac{M_{ag_{obj}}}{l_{px}} \quad (4.14)$$

One way to make this trade-off is to express s_{min} in pixels via Equation 4.15, which combines the pixel density with the spatial blur. Dividing the sensor dimension by this quantity gives the ratio SR (Equation 4.16), which is equivalent to $L_{FoV_{obj}}/s_{min}$ and therefore represents the number of resolvable features across the object field of view. Specifying SR thus fixes how finely the flow is sampled, and with it the number of pixels available to image each feature. For example, $s_{min[px]} = 10$ px ($SR \approx 200$) could be selected: the feature is then spanned by 2–3 speckles and correlated with a 32 px interrogation window at 75% overlap, giving a vector spacing of 8 px and ensuring the feature is resolved by the vector field.

$$s_{min[px]} = s_{min} \cdot \rho_{px_{obj}} \quad (4.15)$$

$$SR = \frac{L_{sens}}{s_{min[px]} \cdot l_{px}} \quad (4.16)$$

By defining SR , an isoline can be drawn in the surface maps where that condition is met. Together with the previous recommendations: select the highest f for robustness, highest $f_{\#}$ to obtain the better S/s_{min} trade-off, while keeping Δ_s small enough; the physical distances l , m and d_f can be defined. This way, a design point can be selected that optimizes the performance parameters given the available resources.

4.5. Summary of Design Guidelines

Taken together, the geometric analysis of Section 4.1, the discussion of Section 4.2, the design space exploration of Section 4.3 and the SR design criterion of Section 4.4 establish a consistent set of guidelines for both BOS and SBOS setups:

- Higher focal lengths make the SBOS system more robust to positional errors and increases the sensitivity limit for BOS. If size is a constraint, a lower f can result in a more compact setup
- A higher f-stop number is desirable, however, it should be such that the an ideal speckle size of 2-5 pixels is still achieved
- Negatively defocused configuration favors a lower defocus distance to exploit its inherent sensitivity advantage over standard BOS

- If f and $f_{\#}$ are maximized, the SR criterion can be used to select the setup that gives the best balance between the optical and CC spatial resolution
- SBOS allows for finer speckle sizes than printed BOS, making it possible to apply sBOS to highly magnified problem ($M_{ag} \approx 1$).

Taken together, these design guidelines constitute the answer to the first research sub-question posed in Chapter 1, namely what the practical guidelines for the design of a SBOS setup should be when balancing the trade-off between sensitivity and spatial resolution. They were used to select the design point of the first experimental campaign, balancing sensitivity and spatial resolution. The following chapter describes the resulting experimental setups, built as a proof of concept to validate this design strategy before the method is applied to the more demanding wind tunnel environment.

5

Experimental Setups

A small experimental setup using a simple object as a case study was built to prove the presented concepts and validate the design strategy. Moreover, this "proof of concept" experimental campaign was used to compare multiple optical configurations, assess their performance and investigate the underlying uncertainty sources. Later, a second experiment was conducted to improve the speckle pattern quality and image a compressible jet. All the information gathered was then applied to the visualization of the flow through an aerospike nozzle in a supersonic wind tunnel. The description of each setup follows below.

Each campaign is described by the same structure: the objective and the research question it addresses, the object of study, the optical layout and hardware, definition of the design points, the acquisition and processing settings, and the test matrix. The uncertainty sources of each campaign are treated in Chapter 6, alongside the results they affect.

5.1. Angled Glass Experiments

The first experimental campaign was envisioned to validate the surface maps presented in the last section and effectively compare a negative and positive defocus configuration in terms of its performance parameters for a similar object field of view. In order to achieve that, an experimental setup was built where the optical distances could be easily set and modified. The setup was assembled on an optical rail close to the ground to avoid vibrations and followed the same structure as the schematic shown in Figure 4.2.

Regarding hardware, Figure 5.1 shows the assembled setup. The laser used was a LIMAB model RC 0.5, a class 2 continuous red light laser ($\lambda = 632.8 \text{ nm}$) with 0.5 mW of power. Given the class of the laser, safety was inherently guaranteed against accidental exposure, as the risk of eye injury is limited by the blink reflex. Nevertheless, simple procedures were made to make sure that the laser paths were contained and terminated. To diffuse the laser and create the speckle pattern, a small ground glass was used with only 1.27 cm of diameter. As it can be seen in the picture, the ground glass was positioned close to laser collimator, which was adjusted to fill the whole diameter of the ground glass. To image the displacements, LaVision's Imager sCMOS CLHS was used, it has a pixel size of $6.5 \mu\text{m}$ and a sensor dimension of 2560×2160 pixels. The camera was equipped mainly with a 200 mm camera lens. Two images with a 105 mm lens were taken, however, it was quickly changed back to the 200mm one, as it provided a more stable and robust assembly.

To simplify the experiment, a small angled piece of glass was used as refractive object, similar to what Schwarz & Braukmann [44] and Meier & Rösger [32] have used. As shown in Figure 5.2, the angled glass introduces a uniform unidirectional displacement to the speckle background that can be predicted using Equation 5.1. By dividing the displacement measured by the camera Δx and the displacement induced by the refractive object ε , it is possible to obtain the setup sensitivity S , as shown in Equation 4.4. It is worth noting that the definition of ε in this case is an artifice because the real displacement Δx is independent of l . The ε is used to connect a simple refractive object to a real experimental scenario.

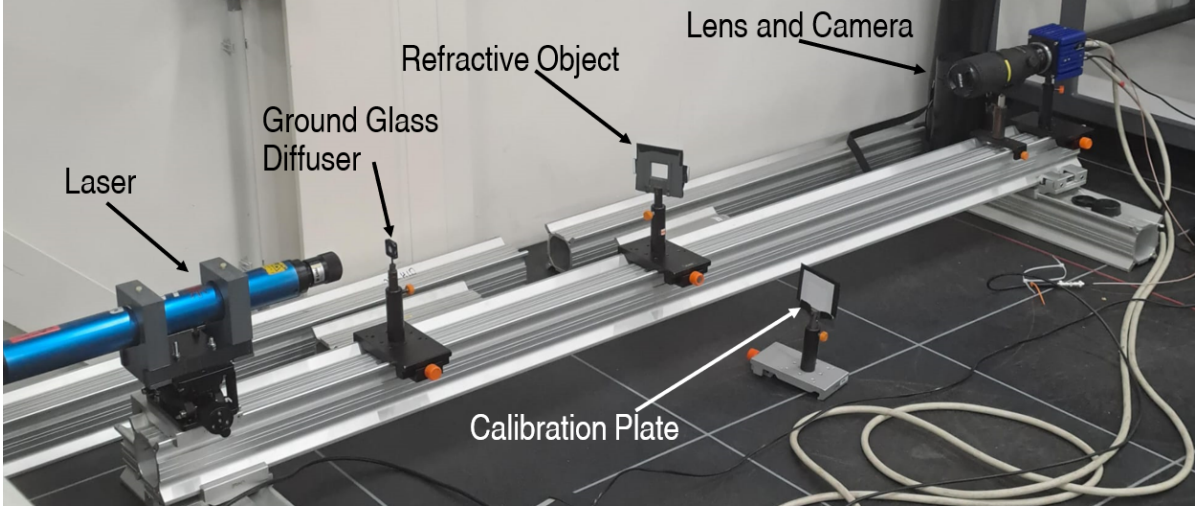


Figure 5.1: Assembled setup of the angled glass campaign. The laser and the ground glass diffuser are mounted at the near end of the rail, the angled glass refractive object at mid-span and the lens and camera at the far end. The calibration plate, used to measure the optical magnification M_{ag} in the in-focus plane, is shown on its own mount.

The glass piece used had a thickness $T = 2$ mm and a refractive index of around $n = 1.52 \pm 0.02$.

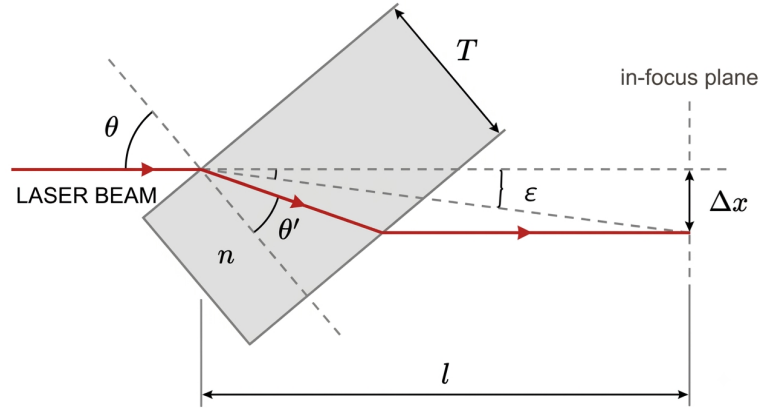


Figure 5.2: Geometry of the angled glass plate used as refractive object. A beam incident at an angle θ on a plate of thickness T and refractive index n is refracted to θ' and leaves the plate parallel to its original direction, laterally displaced by Δx . Over the defocus distance l this displacement is equivalent to a uniform deflection angle ε .

$$\varepsilon = \frac{\Delta x}{l} = \frac{T}{l} \sin(\theta) \left(1 - \frac{\cos(\theta)}{\sqrt{n^2 - \sin^2(\theta)}} \right) \quad (5.1)$$

Table 5.1 shows the information of each point tested during the first campaign. The table lists not only the camera, lens configuration and optical distances, but also the theoretical magnification, sensitivity, predicted displacement given the measured glass angle and whether the in-focus plane was behind or in front of the refractive object. The points vary the type of configuration, the f-stop number, the defocus distance and the focal length of the camera and all of them have been obtained by enforcing the object field of view to be 1.7cm and inspecting the surface maps obtained with the code described in Chapter 4. In this instance then, the SR criterion was not used, as the intention of the experiment was to validate the methodology by testing various different setups. Therefore, by comparing the theoretical sensitivity, magnification and the displacement with the ones obtained experimentally at each of these points, the methodology presented in Chapter 4 can be validated. Moreover, through the comparison

of experimental sensitivities obtained with a negative defocus against a positive defocus configuration, the difference in performance of SBOS versus BOS is obtained.

Table 5.1: Optical configurations acquired during the angled glass campaign. θ is the angle at which the glass plate was set. "Negative" and "Positive" denote the in-focus plane lying respectively in front of or behind the test section, following the convention of Chapter 4. Points 1-9 correspond to the experiment with a similar L_{FOV} , whereas points 10-20 correspond to the linear slider experiment. In point 10 the refractive object is coincident with the in-focus plane, so no defocus type is defined.

Point	$f_{\#}$	f [mm]	l [mm]	d_f [mm]	M_{ag} [-]	S [mm]	θ [°]	Type of Defocus
1	16	200	80	409.0	0.956	76.50	17	Negative
2	16	200	80	505.0	0.656	52.50	17	Positive
3	16	200	100	391.0	1.050	105.00	17	Negative
4	16	200	100	511.0	0.643	64.30	17	Positive
5	11	200	80	393.0	1.037	83.00	17	Negative
6	11	200	80	490.0	0.690	55.20	17	Positive
7	11	105	80	189.0	1.250	100.00	20	Negative
8	4	105	80	285.0	0.583	46.60	20	Positive
9	16	200	150	527.0	0.611	91.60	20	Positive
10	8	200	0	470.0	0.741	0.00	20	–
11	8	200	10	470.0	0.741	7.41	20	Positive
12	8	200	20	470.0	0.741	14.81	20	Positive
13	8	200	30	470.0	0.741	22.22	20	Positive
14	8	200	40	470.0	0.741	29.63	20	Positive
15	8	200	50	470.0	0.741	37.04	20	Positive
16	8	200	10	470.0	0.741	7.41	20	Negative
17	8	200	20	470.0	0.741	14.81	20	Negative
18	8	200	30	470.0	0.741	22.22	20	Negative
19	8	200	40	470.0	0.741	29.63	20	Negative
20	8	200	50	470.0	0.741	37.04	20	Negative

Points 10-20 of Table 5.1 describe the optical arrangement of a different configuration to the previous points. The setup presented in Figure 5.1 was modified to run an experiment intended to demonstrate the SBOS capacity of achieving different defocus configurations. The experiment consisted in keeping the in-focus plane fixed and moving the refractive object in steps with a linear slider, shown in Figure 5.3. This kept the optical magnification constant, while permitting the exploration of a range of defocusing distances. Therefore the experiment can establish whether the advantage of the negative defocus configuration in achieving a higher sensitivity comes only from M_{ag} , or also from some other mechanism related to l .

The data acquisition followed standard BOS procedure using LaVision's DaVis 8.4 software. First, a calibration plate was used to define the in-focus plane, it can be seen in Figure 5.1. From the calibration, DaVis computed the magnification and the pixel density of the setup. Then, the laser is turned on and a reference image is taken. The laser is turned back off and the refractive object is put into position following the distances presented in the surface maps of Chapter 4. With the laser back on, a displaced image is taken. Post-processing was done using LaVision DaVis 11 software. The images were first pre-processed to obtain a 2 frame sequence. Then the cross-correlation (CC) algorithm was used to reconstruct the displacement. Due to an unexpected large average speckle size, the smallest interrogation window had a size of 64×64 with 4 multipass and 75% of overlap, while the biggest had 256×256 pixels with 1 pass and 75% overlap. Nevertheless, given the large magnification, the vector pitch obtained is still quite small.

Another aspect of the experimental setup was that no beam expander was used during the first experimental campaign. Since no specification in the literature was found to size the laser diffuser and given the small field of view of 1.7cm, it was believed that the scattering and diffusion alone from the ground glass would be sufficient to fill the whole field of view. However that belief was proven wrong. Moreover, the small size of the diffuser made the laser beam diffraction limited at the ground glass and not at the

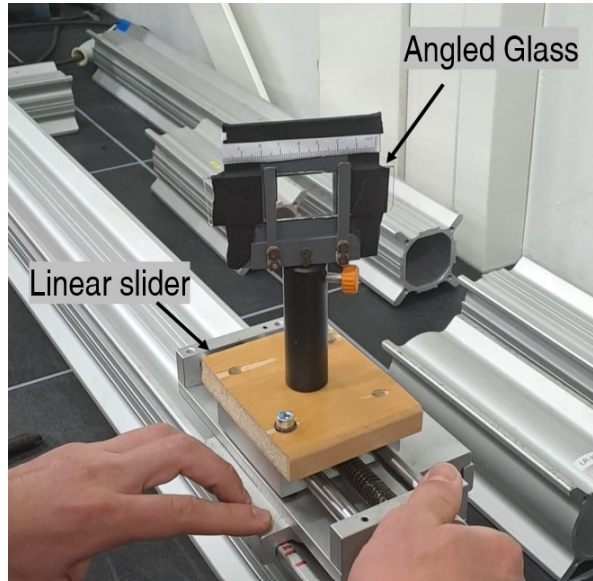


Figure 5.3: Picture of the angled glass in a linear slider, used to investigate the effect of M_{ag} in the sensitivity

camera aperture, this meant that changing the f-stop of the camera had no effect in the average speckle pattern size and the average speckles were a lot bigger than expected. Another effect was a change in the speckle pattern diameter with the aperture diameter and magnification ($D_{\Delta s} \approx D \cdot M_{ag}$), which limited the usable $f_{\#}$. Faced with these limitations, another experimental campaign was envisioned with a bigger ground glass and a beam expander to try and produce a finer speckle pattern.

5.2. Speckle Pattern Investigation

The second experimental campaign was intended to investigate further the speckle pattern and tackle the limitations of the first experiments. The main goals were to better understand the creation of a finer speckle pattern assess how the average speckle size correlates to the Rayleigh limit of Equation 4.9 and also compare the generation of speckles through direct ground glass diffusion and by projecting the laser into a surface. The setup used for the direct speckles investigation resembles the one shown in Section 5.1 and can be seen in Figure 5.4.

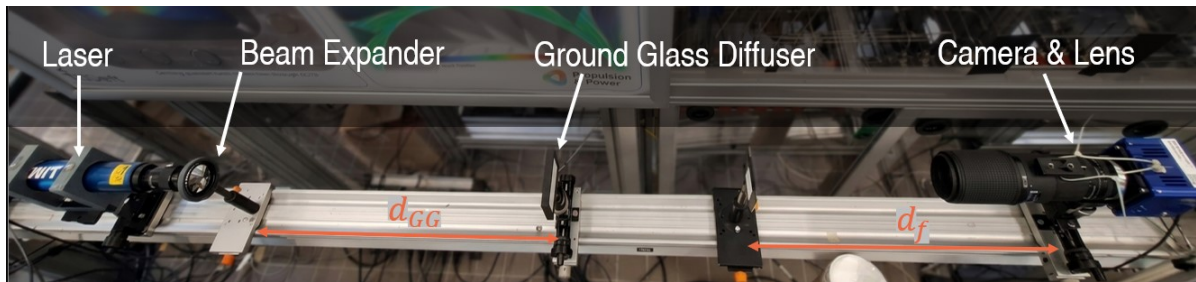


Figure 5.4: Assembled directed speckle investigation setup. Configuration is quite similar from the one presented in Section 5.1 with the distinct presence of the beam expander lens in-between the laser and ground glass diffuser.

The main difference between the angled glass setup, in Figure 5.1, and the one used for speckle investigation, in Figure 5.4, is the presence of a larger ground glass diffuser and a lens to expand the laser beam. The new ground glass diffuser has 88 mm of length and 61 mm of width, which makes it capable of filling larger fields of view. The beam expander consists of a convergent lens with focal length of $f = 80$ mm, positioned at a distance d_{GG} behind the ground glass, which was varied to see its effects in the speckle creation. The camera and laser are the same from the last experiment, only the camera lens with $f = 200$ mm was used. One detail that was identified in this campaign is the fact that the $f_{\#}$ markings in the camera lens correspond to rounded members of a geometric series of ratio

$\sqrt{2}$. That introduces a error of around 2.9% for the points where an $f_{\#}$ of 11 and 22 has been used, therefore it has been decided to report the actual value rather than the marked one here.

Table 5.2: Configurations acquired during the speckle pattern investigation. The f-stop number is symbolized by $f_{\#}$, d_{GG} is the distance between the beam expander lens and the ground glass diffuser, M_{ag} the optical magnification and Δs_{th} the speckle size predicted by the Rayleigh diffraction limit (Equation 4.9).

Point	$f_{\#}$ [-]	d_{GG} [mm]	M_{ag} [-]	d_f [mm]	Δs_{th} [px]
1	4.00	740	0.9302	415	0.92
2	11.31	740	0.9302	415	2.59
3	22.63	740	0.9302	415	5.19
4	4.00	740	0.2740	930	0.61
5	11.31	740	0.2740	930	1.71
6	22.63	740	0.2740	930	3.42
7	11.31	450	0.2740	930	1.71
8	11.31	280	0.2740	930	1.71
9	11.31	190	0.2740	930	1.71
10	11.31	100	0.2740	930	1.71
11	4.00	300	0.8511	435	0.88
12	11.31	300	0.8511	435	2.49
13	22.63	300	0.8511	435	4.97
14	22.63	300	1.0000	400	5.37
15	22.63	300	0.9091	420	5.13
16	22.63	300	0.8000	450	4.84
17	22.63	300	0.6667	500	4.48
18	22.63	300	0.5714	550	4.22
19	22.63	300	0.5000	600	4.03
20	22.63	300	0.4000	700	3.76
21	22.63	300	0.3333	800	3.58
22	22.63	300	0.2857	900	3.46
23	22.63	300	0.2222	1100	3.28
24	22.63	300	0.1667	1400	3.14
25	22.63	300	0.1176	1900	3.00
26	22.63	300	0.0741	2900	2.89
27	22.63	300	0.0541	3900	2.83
28	22.63	300	0.0351	5900	2.78

A series of pictures were taken where d_{GG} , M_{ag} and $f_{\#}$ were varied. Table 5.2 summarizes the experimental condition of each picture. Points 1-3 correspond to a f-stop sweep maintaining the same d_{GG} and M_{ag} and points 4-6 did the same for a different M_{ag} . Then Points 7-9 changed d_{GG} with a fixed f-stop and M_{ag} . The first block ends with another f-stop sweep with different d_{GG} and M_{ag} at points 10-13. The second block, from point 14 to 28, fixed the f-stop and d_{GG} to vary the optical magnification following the lens markings. This considerably reduced experimental time, as each different d_f was not verified with a calibration plate, however, it lost accuracy as it was only based on the marking of the lens. For each point, the average speckle size was computed through an autocorrelation of the intensity as described by Hamarová et al. [18] and using LaVision's DaVis 11 software processing function *particle/speckle size over time* which calculates the spatial averaged particle size through a defined sliding average length and subtracting the sliding minimum [38]. For this application, both quantities were set to 16 pixels. These pictures will enable the study of a possible ground glass distance capable of producing the desired speckle pattern and quantify the effects of the magnification and f-stop number in the speckle production.

DaVis 11 documentation is quite obscure on the method used to compute the average speckle size, citing only Pan et al. which concerns a way to quantify speckle quality. In the other hand, Hamarová et al. [18] describes clearly a method to compute the average speckle size by fitting an autocorrelation func-

tion to intensity data. The image is transformed into a one-dimensional vector and an autocorrelation function is applied to it. the mean speckle radii is then computed as the shift at which the normalized peak decreases to $1/e$. This definition corresponds to a half width of the correlation peak, therefore the average speckle size will be the full width. As an standard practice in speckle imaging [47, 49], a pre-processing step must be done before applying Hamarová et al.'s method. The recorded intensity does not contain the speckle fluctuation alone, it also presents an underlying illumination envelope from directly imaging the beam. This added intensity is correlated over hundreds of pixels and biases the measured size towards larger values. In order to avoid it, each image was normalized by a Gaussian low-pass filtered copy of itself before the correlation. The width of the filter was selected such that the autocorrelation tail vanishes beyond the correlation peak.

The previous setup was modified to also test the creation of speckles by projecting the laser in a blank canvas, similar to what was done by Meier & Rösigen and Bühlmann [32, 4]. As shown by the schematic in Figure 5.5a, a blank surface angled at 45° has been placed in the beam path and the camera assembly has been moved to a perpendicular beam. The laser is expanded by the same $f = 80$ mm lens. When it hits the surface, its light diffuses and creates an interference pattern captured by the camera. Figure 5.5b shows a picture of the speckle pattern being generated in the projection plane, there it is also possible to see the camera assembly and the beam expander.

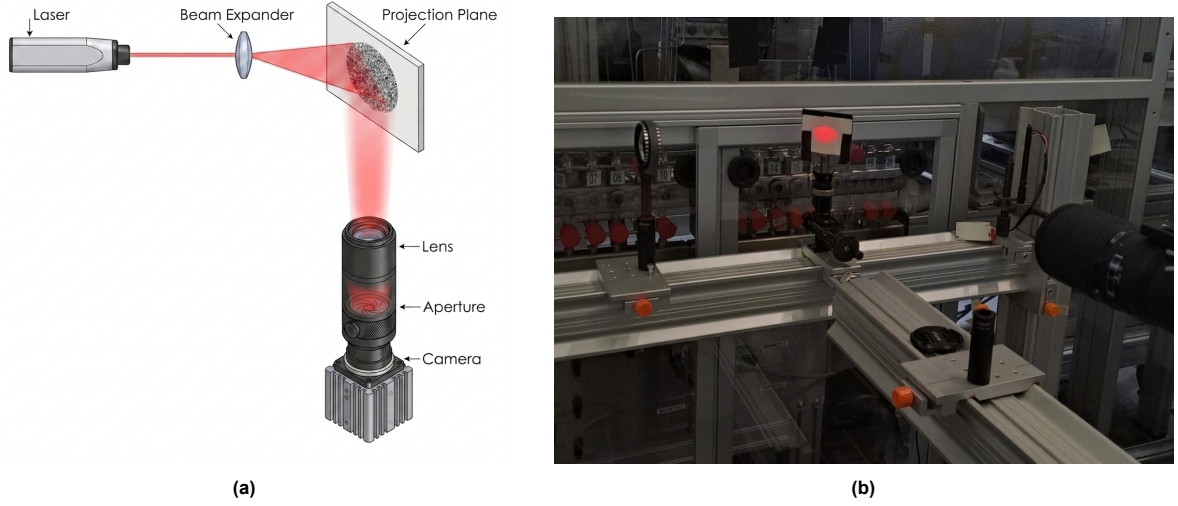


Figure 5.5: Setup used to create and study projected speckles. (a) Diagram of the arrangement. (b) Photograph of the assembled setup, where the speckle pattern formed in the projection plane is visible together with the camera assembly and the beam expander.

3 points were obtained using projected speckles and they are summarized in Table 5.3 as points 29-31. A $f_{\#}$ sweep was done to assess the effect of that variable with a d_{GG} of 300mm, as it was the distance where the most amount of points were collected for the directed speckle case. Therefore, a comparison can be done between the results obtained with projected speckles and direct speckles.

Table 5.3: Configurations acquired with projected speckles. The columns follow the convention of Table 5.2 and the point numbering continues from it, $f_{\#}$ is the f-stop number, d_{GG} is the distance between the beam expander lens and the projection plane, M_{ag} the optical magnification and Δs_{th} the speckle size predicted by the Rayleigh diffraction limit (Equation 4.9). The three points share the same optical layout and vary only the aperture.

Point	$f_{\#}$ [-]	d_{GG} [mm]	M_{ag} [-]	d_f [mm]	Δs_{th} [px]
29	4.00	300	0.4167	680	0.67
30	11.31	300	0.4167	680	1.90
31	22.63	300	0.4167	680	3.81

With the data gathered, the setups were tested on how to create an appropriate speckle pattern and how to set the optical distances so a desirable sensitivity and spatial resolution is achieved. This

knowledge now can be applied to a study case that enables the visualization of a true density gradient and even its integration into a density field. The next experiment will be build on that basis to visualize a compressible jet, which will serve as proof of concept and allow for the reconstruction of a density field using SBOS.

5.3. Compressible Jet Experiment

The goal of this experiment was to image a real compressible flow using SBOS, which could later be post-processed into quantitative measurements. For that, a similar setup to the one shown in Figure 5.4 was used to image a compressible underexpanded jet. This experiment used $l = 3$ cm, $d_{GG} = 83$ cm, $d_f = 74$ cm and $f_{\#} = 11$, which gives $M_{ag} = 0.37$, $S = 11.1$ mm, $s_{min} = 0.369$ mm and a $L_{FoV_{obj}} = 41, 5 \times 35$ mm. Figure 5.6 shows a picture of the assembled setup. The jet was achieved by connecting a standard air gun with a nozzle diameter $d_N = 3.5$ mm to a wall air tap with a hose. This setup allowed to adjust the gauge pressure coming out of the air tap. Initial runs experienced considerable issues with the vibrations that came with the activation of the jet, so a solution was implemented where the air gun had to be mounted on a different rail transversal to the one where the optical equipment was, this can be seen in Figure 5.6. A big limitation of the experiment, however, was the lack of loss quantification through the hose, quick-disconnect couplings and the air gun itself, meaning that the actual pressure right before the nozzle exit was unknown.

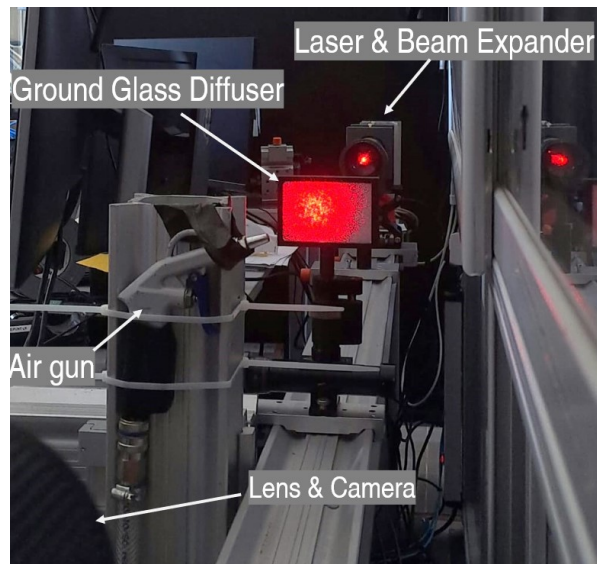


Figure 5.6: Photograph of the compressible jet experimental setup.

The data acquisition followed a similar procedure than the one explained at Section 5.1. A target plate was put at the in-focus position and a picture was captured to calibrate laVision's DaVis software. Then, the laser was turned on and a reference picture was taken. The pressure was set at the air tap valve and the air gun was triggered, while the camera captured 100 images with an exposure time of 300 μ s. In total, 6 cases were captured with pressures at the air tap valve ranging from 3 bar to 8 bar with 1 bar step. The speckle size obtained was checked following the procedure of Section 5.2. The 100 images for each case were averaged and cross-correlated with an interrogation window of 12 px, 75% of overlap and 4 multipasses where the maximum window was set to 256x256 px.

5.4. Aerospike Experiment

6

Results

Write Introduction paragraph

6.1. Angled Glass Experiment Results

The Angled Glass Experiment aimed to validate the methodology and compare the different defocus configurations treated in Chapter 4. This section presents the data gathered, describes how it was treated, discusses the uncertainty of the results, and concludes whether these objectives have been achieved. The tables that follow give an overview of the results.

Table 6.1 compares the displacements measured through SBOS with the predicted shift induced by the angled glass. Δ_x is the mean of the cross-correlation vector field in pixels, Δx_{SBOS} is the latter transferred to mm in the focal plane and Δx_{glass} is the glass-induced ray shift, calculated through Equation 5.1. Since the glass piece produces a uniform displacement field, the standard deviation of the cross-correlation field σ_Δ quantifies the precision of the SBOS measurement, while the relative error quantifies the accuracy of the method. The last column displays the total combined uncertainty of measurement of Δx , which will be treated in more detail later this section.

The table allows the comparison of the measurement error with its inherent uncertainty. In general terms, the error obtained is below the combined uncertainty, with the exception of points 1, 17 and 18. Only point 1, however, differs significantly from the uncertainty, with a difference of 4.76 percentage points, while 17 and 18 deviate by less than 0.5 percentage points. Regarding point 1, an error of 1.26° in the glass angle measurement would be needed to cause its deviation of 7.87%, which exceeds the reading uncertainty of the protractor. The cause for such error could be a misread angle or an unnoticed movement that changed the angle when setting it in the rail.

The slider experimental error is also worth a remark due to the distinct behavior of its deviation. As it can be seen in Table 6.1, all the deviations of the slider experiments are positive and have a similar magnitude, which leads to the interpretation of an induced bias in the measurement. It is not clear what caused this bias, although it could have been introduced by an error in the magnification added to one in the glass angle, as both were fixed throughout the measurements. Nevertheless, the deviation of every point except 1 stays well within 5%, below its measurement uncertainties, and the scatter signaled by σ_Δ is well within 2px. Consequently, these metrics indicate that the technique has been well implemented and is capable of producing accurate and precise results.

Through Table 6.2, the design strategy of Chapter 4 can be assessed. It presents a comparison of the optical distances and magnifications prescribed by the surface maps with the ones obtained through the focal plane calibration. In addition, the resulting L_{FoVobj} was compared with the enforced 17 mm field of view of the object. The table was restricted to the 9 points where the methodology of Chapter 4 was used to size the optical setup. The relative magnification errors are all below 5% with the exception of point 7. The latter was obtained using the minimum distance at which the calibration plate could be brought into focus when using an $f = 105$ mm lens. That distance was bigger than the one prescribed

Table 6.1: Displacements measured in the angled glass campaigns. Δ_x is the mean displacement returned by the cross-correlation, σ_Δ its standard deviation over the vector field, Δx_{SBOS} is that displacement referred to the focal plane, Δx_{glass} the value calculated by Equation 5.1 for a plate of thickness $T = 2$ mm and refractive index $n = 1.52$ at the measured angle θ . The last column is the relative deviation of the two. Points 1 to 9 follow the fixed field of view configurations of Table 5.1; points 10 to 20 follow the linear slider configurations of the same table. The column u_{tot} gives the combined standard uncertainty of the comparison between Δx_{SBOS} and Δx_{glass} calculated from the budgets of Tables 6.4 and 6.5.

Point	θ [°]	Δ_x [px]	σ_Δ [px]	Δx_{SBOS} [mm]	Δx_{glass} [mm]	Δx_{SBOS} dev. [%]	u_{tot} [%]
1	17	29.818	0.6001	0.1933	0.2099	-7.87	3.11
2	17	22.613	1.3660	0.2136	0.2099	+1.77	6.49
3	17	34.355	0.8521	0.2137	0.2099	+1.84	3.43
4	17	21.368	0.5763	0.2071	0.2099	-1.30	3.59
5	17	33.414	0.7710	0.2082	0.2099	-0.78	3.31
6	17	23.067	1.0230	0.2088	0.2099	-0.51	5.03
7	20	41.386	1.8830	0.2552	0.2500	+2.05	5.05
8	20	22.306	0.6617	0.2537	0.2500	+1.46	3.68
9	20	25.238	0.7720	0.2565	0.2500	+2.59	3.76
10	20	29.373	0.7733	0.2561	0.2500	+2.42	3.42
11	20	29.432	0.8055	0.2566	0.2500	+2.63	3.50
12	20	29.595	0.7575	0.2580	0.2500	+3.20	3.36
13	20	29.613	0.7711	0.2582	0.2500	+3.26	3.40
14	20	29.580	0.7661	0.2579	0.2500	+3.15	3.39
15	20	29.662	0.8175	0.2586	0.2500	+3.43	3.51
16	20	29.559	0.7815	0.2577	0.2500	+3.07	3.43
17	20	29.653	0.7694	0.2585	0.2500	+3.40	3.39
18	20	29.746	0.7815	0.2593	0.2500	+3.73	3.41
19	20	29.653	0.8264	0.2585	0.2500	+3.40	3.54
20	20	29.617	0.8570	0.2582	0.2500	+3.28	3.62

by the surface map, resulting in a large d_f , M_{ag} and L_{FoVobj} error. The propagation of that error is also evident in the sensitivity shown in Table 6.3. Point 9 also deviates from the enforced field of view by more than 5%, in this case by -8.64%. It was acquired at $l = 150$ mm, the longest defocus distance of the campaign. Points 7 and 9 are therefore the two geometric extremes of the design space, the shortest focal length and the largest defocus, and they are also the only two points whose field of view falls outside 5%. All remaining deviations are within that band, which signals the correct implementation of the method. The contribution of the optical magnification to the uncertainty was obtained through the maximum fit residual of the DaVis calibration and can be seen in Table 6.5.

The data collected during the experimental campaign allows the assessment of sensitivity through two experimental routes. Table 6.3 compares the theoretical sensitivity given by the surface map of Chapter 4 to the sensitivity obtained through the geometry $S_{geom} = l \cdot M_{agexp}$ and the sensitivity obtained with the displacement data $S_\Delta = \Delta_x \cdot l_{px} / \varepsilon_{glass}$. With the exception of point 7, which has already been discussed, all data is within 8% of agreement. It is worth noting that the S_{geom} error is the same as the magnification error, while S_Δ stacks both the displacement measurement error and the geometric setup error. The total combined uncertainty of S_Δ has also been calculated and is displayed in the last column, most points are below u_{tot} in magnitude or within 1% to it with the exception of point 7 and 9. It is worth noting that most points which error exceeds the total uncertainty belong to the slider experiment, which could have experienced a bias, as previously mentioned. Nevertheless, The data shows that the optical configuration parameters defined by the methodology of Chapter 4 produce a setup with the predicted sensitivity within an acceptable error range. Therefore, it validates the methodology as it demonstrates that the physical setup reproduces the predicted performance.

Tables 6.1 and 6.3 both show total uncertainties that were obtained to better assess the measurement error. These values were computed through uncertainty budgets following the Guide to the Expression of Uncertainty in Measurement [26]. The campaign compared a displacement obtained by Equation

Table 6.2: Optical design values against measured values of the experimental campaign. $d_{f,th}$ and $M_{ag,th}$ are, respectively, the focal distance and optical magnification prescribed by the design methodology of Chapter 4 and listed in Table 5.1. $M_{ag,exp}$ was measured with a calibration plate at the in-focus plane and $d_{f,exp}$ was calculated through $d_f = f(M_{ag,exp} + 1)/M_{ag,exp}$. The d_f and M_{ag} deviation columns are the relative difference with respect to the design value. L_{FoVobj} is the field of view achieved at the refractive object, obtained from Equation 4.8 with the experimental magnification. Its deviation was taken with respect to the 17 mm L_{FoVobj} enforced during the design.

Point	$d_{f,th}$ [mm]	$d_{f,exp}$ [mm]	d_f dev. [%]	$M_{ag,th}$ [-]	$M_{ag,exp}$ [-]	M_{ag} dev. [%]	L_{FoVobj} [mm]	L_{FoVobj} dev. [%]
1	409.0	399.5	-2.32	0.9561	1.0025	+4.85	16.81	-1.12
2	505.0	490.6	-2.85	0.6557	0.6882	+4.96	17.07	+0.43
3	391.0	391.4	+0.11	1.0498	1.0449	-0.47	16.87	-0.76
4	511.0	498.2	-2.50	0.6433	0.6706	+4.24	16.73	-1.56
5	393.0	391.7	-0.32	1.0369	1.0431	+0.59	16.21	-4.65
6	490.0	478.5	-2.35	0.6897	0.7181	+4.12	16.28	-4.22
7	189.0	204.6	+8.25	1.2500	1.0543	-15.66	18.52	+8.97
8	285.0	288.7	+1.30	0.5831	0.5715	-1.98	17.76	+4.46
9	527.0	512.7	-2.71	0.6109	0.6395	+4.69	15.53	-8.64

5.1 against the displacement measured by cross-correlation. So two budgets were constructed, one for the glass displacement and another for the CC measurement. Afterwards, the collected uncertainties were combined to obtain the sensitivity uncertainty. A description of each budget and the formulas used to assess the total contribution follows.

The glass displacement's uncertainty budget is summarized in Table 6.4. It focuses on the three inputs of the angled glass displacement estimate, namely the plate thickness T , the plate angle θ and the refractive index n . Thickness was measured with a caliper and the angle was set with a protractor, while the refractive index was obtained from the literature for common glass. Each input was treated as a rectangular probability distribution of half-width a with a standard uncertainty of $u = a/\sqrt{3}$. The sensitivity coefficients and contributions of each input variable regarding the two glass angles were also estimated based on Equation 5.1. These coefficients measure how much a change in each variable affects the final displacement. They are also used to calculate the contribution of each variable to the combined uncertainty as it is equal to the sensitivity coefficient times uncertainty [26]. These quantities are presented in Table 6.4, where the half-width and the standard uncertainty are listed for every input, followed by its contribution at each glass angle. The combined uncertainty for each angle is given at the bottom row and corresponds to the root of the sum of squares of the above percentages.

Regarding the uncertainty of the cross-correlation displacement measurement, it is affected mainly by two variables. The first is the scatter of the cross-correlation vector field σ_Δ , listed in Table 6.1. It ranges from 2.0% to 6.0% of the mean displacement with an average of 3.0%. This term is evaluated from the vector field itself, varying from point to point. The second is the optical magnification returned by the DaVis calibration. The calibration fit residual stayed within 2.9 px over the sensor width of 2560 px, which gives a contribution of 0.11%. The pixel pitch l_{px} is fixed by the manufacturer and is negligible against the other two. Table 6.5 presents the second budget, together with the uncertainty of the defocus distance l , which is required for the sensitivity.

The total uncertainties that follow from the budgets are reported per point in the last column of their respective tables. The first total concerns the comparison between the predicted and the measured displacement. It combines the prediction budget with the scatter and the magnification, and it ranges from 3.11% at point 1 to 6.49% at point 2. The second total concerns the sensitivity S_Δ , which stays within 3.19% to 6.75%. It reaches its largest values at points 11 and 16, for which the defocus distance of only 10 mm makes the contribution of l dominant. Points 2, 6 and 7 sit above 5% in both totals. In each case the cross-correlation scatter is the dominant contribution.

Regarding the research objectives, one of the goals of the experimental campaign was the comparison of the new capacities of SBOS to a standard BOS configuration. With the data at hand it is possible to compare the performance of a setup that uses negative defocus to another with positive defocus, effectively answering the experiment's objective. Figure 6.1a plots the sensitivity versus defocus distance

Table 6.3: Sensitivity obtained by two routes compared to theoretical sensitivity. $S_{th} = lM_{ag_{th}}$ is the design sensitivity of Table 5.1. $S_{geom} = lM_{ag_{exp}}$ is the sensitivity measured with the calibration plate, carrying the same deviation of Table 6.2. $S_{\Delta} = \Delta_x \cdot l_{px} / \epsilon_{glass}$ follows from the displacements measured. The last two columns give the deviation of S_{geom} and S_{Δ} to S_{th} . The column u_{tot} displays the combined standard uncertainty on S_{Δ} for each point, obtained from the budgets of Tables 6.4 and 6.5.

Point	S_{th} [mm]	S_{geom} [mm]	S_{Δ} [mm]	S_{geom} dev. [%]	S_{Δ} dev. [%]	u_{tot} [%]
1	76.5	80.2	73.9	+4.84	-3.42	3.19
2	52.5	55.1	56.0	+4.87	+6.73	6.53
3	105.0	104.5	106.4	-0.49	+1.34	3.48
4	64.3	67.1	66.2	+4.29	+2.93	3.64
5	83.0	83.4	82.8	+0.54	-0.24	3.39
6	55.2	57.4	57.2	+4.08	+3.55	5.08
7	100.0	84.3	86.1	-15.66	-13.93	5.10
8	46.6	45.7	46.4	-1.88	-0.45	3.75
9	91.6	95.9	98.4	+4.73	+7.44	3.78
10	0.0	0.0	0.0	-	-	-
11	7.4	7.5	7.7	+0.75	+3.40	6.75
12	14.8	14.9	15.4	+0.75	+3.97	4.43
13	22.2	22.4	23.1	+0.75	+4.04	3.90
14	29.6	29.8	30.8	+0.75	+3.92	3.68
15	37.0	37.3	38.6	+0.75	+4.21	3.70
16	7.4	7.5	7.7	+0.75	+3.85	6.71
17	14.8	14.9	15.4	+0.75	+4.18	4.45
18	22.2	22.4	23.2	+0.75	+4.50	3.92
19	29.6	29.8	30.8	+0.75	+4.18	3.82
20	37.0	37.3	38.5	+0.75	+4.05	3.80

of points 1 to 9 that were captured with $f = 200$ mm, with error bars corresponding to the standard deviation divided by the mean cross-correlation displacement times the sensitivity values.

Figure 6.1a replicates one of the predictions of Chapter 4, where a higher sensitivity was obtained for the same defocus distance if a negative defocus was chosen instead of its positive one. It is also worth noting that all points were obtained with the same field of view of the object. Therefore, a higher sensitivity and magnification is achieved while capturing the same area of the object. Although the resolution has not been measured in this case, it is also to be expected through Equation 4.12 that the spatial resolution trade-off is improved by this increase in magnification. This figure effectively compares the performance of SBOS with BOS, answering one of the research questions posed in Chapter 1.

Points 10 to 20 of Tables 6.1 and 6.3 show the results of the experiment where the angled glass was

Table 6.4: Uncertainty budget for the displacement predicted by Equation 5.1. The three input variables are described under the Source column together with the instrument used to measure them in the instrument column. Each input is treated as a rectangular distribution of half-width a , so that $u = a/\sqrt{3}$ [26]. The contributions c_i are the sensitivity coefficients, computed by inspecting the deviation in Δx_{glass} after a small difference of each variable, multiplied by the uncertainty u . On the bottom row, the combined uncertainty of Δx_{glass} can be found together with the formula used to compute it.

Source	Instrument	a	u	Contribution c_i [%]	
				$\theta = 17^\circ$	$\theta = 20^\circ$
Plate thickness T	Caliper	0.02 mm	0.012 mm	0.58	0.58
Refractive index n	Literature	0.02	0.012	1.40	1.37
Plate angle θ	Protractor	0.50°	0.29°	1.82	1.59
Combined, $u_{glass} = \sqrt{c_T^2 + c_n^2 + c_\theta^2}$				2.37	2.18

Table 6.5: Uncertainty budget for the displacement measured by cross-correlation and sensitivity. The scatter of the vector field is evaluated from the data of Table 6.1 and varies per point. For the magnification, DaVis calibration residual was used as its uncertainty which slightly varies per point, so the maximum was taken here. The pixel pitch is common to the whole campaign. The defocus distance was obtained with a metric tape and judged by about ± 1 mm. The contributions were obtained as explained in Table 6.4. The bottom rows show the combined uncertainty for Δx and S_Δ together with the formulas used to compute them. Since the values vary per point, its range is presented here but they can be found in at Tables 6.1 and 6.3, respectively.

Source	Instrument	a	u	Contribution c_i [%]
Vector field scatter σ_Δ	Cross-correlation	—	per point	2.01 to 6.04
Optical magnification M_{ag}	DaVis calibration	—	2.9 px	0.11
Pixel pitch l_{px}	Manufacturer	—	—	negligible
Defocus distance l	Metric tape	1 mm	0.58 mm	0.38 to 5.77
Combined, $u_{\Delta x} = \sqrt{u_{glass}^2 + \sigma_{rel}^2 + c_{M_{ag}}^2}$				3.11 - 6.49
Combined, $u_{S_\Delta} = \sqrt{u_{glass}^2 + \sigma_{rel}^2 + c_l^2}$				3.19 to 6.75

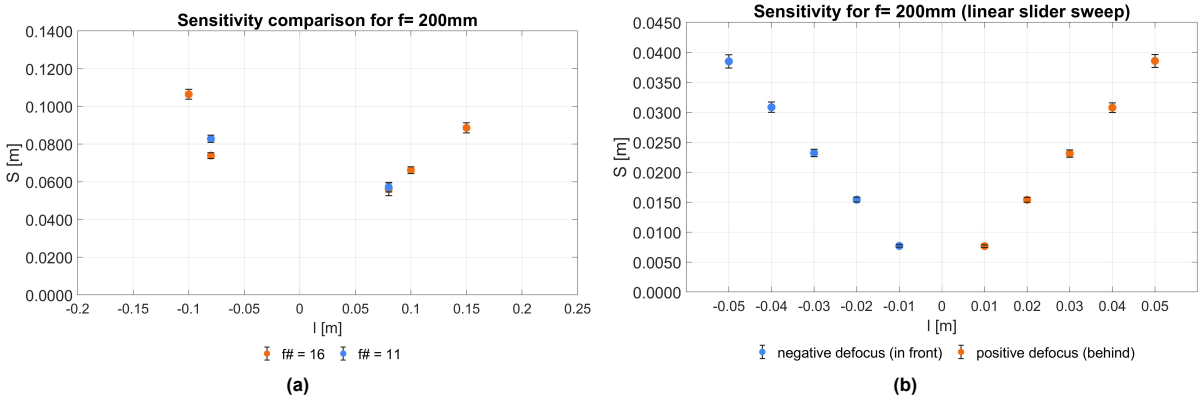


Figure 6.1: Sensitivity versus defocus distance for a camera lens of $f = 200$ mm. (a) Points obtained with $f\# = 11$ and $f\# = 16$ for an enforced $L_{FOVobj} = 17$ mm. (b) Points obtained in the slider experiment, with a fixed measured magnification of $M_{ag} = 0.746$

positioned in a linear slider with a fixed magnification and only the defocus position was changed. Figure 6.1b plots the resulting sensitivity against the defocusing distance. The points are symmetric around the Y-axis, which means that the mechanism that makes a negative defocus configuration more powerful than its positive counterpart is not the negative distance per se but the increased magnification that comes with a shorter focal distance to the camera. Therefore, the mechanism making negative defocus more sensitive than standard defocus is the higher magnification that comes with a shorter focal length.

Section 5.1 mentioned a discrepancy between the predicted average speckle size obtained through the Rayleigh diffraction limit and the ones obtained experimentally and computed using Hamarová's method [18]. Table 6.6 displays both quantities side by side together with the $f\#$ and M_{ag} used at each point. The Δ_s deviation shows that at some points the discrepancy exceeds the theoretical value several times. Speckle formation is an intrinsically random phenomenon, described by complex physics [15, 13, 6], so it is not exactly clear what is precisely the root of this difference. Nevertheless, the data gives some clues on what could be improved.

It seems that the f-stop number did not influence the average speckle size obtained experimentally. This becomes evident from observing points 4 and 8-10. In these points, the $f\#$ ranges from 4 to 16 whereas Δ_s stays within 12.35 px to 12.94 px. In contrast, changes in magnification were observed to have an effect in the experimental average speckle size considerably. From points 1-4 the $f\#$ is fixed and M_{ag} changes from 0.671 to 1.045, displaying Δ_s from 12.74 px to 21.21 px. These factors contradict the Rayleigh equation for subjective speckles where its size is only dependent on the light's wavelength, imaging lens $f\#$ and M_{ag} . The independence to the aperture size indicates that the interference pattern

is not being created at the lens pupil as it should [15]. And the magnitude effect means that the speckle is behaving like an imaged object. These factors lead to the belief that the pattern did not form a Rayleigh distribution at the lens pupil, rather what is being imaged are just the phase cells illuminated at the diffuser.

Table 6.6: Speckle sizes measured during the angled glass campaign, for the configurations of Table 5.1. M_{agexp} is the magnification measured with the calibration plate, Δs_{th} the size predicted by the Rayleigh limit of Equation 4.9 and Δs_{exp} the size returned by the one-dimensional autocorrelation method of Hamarová et al. [18], taken as the lag at which the normalised autocorrelation of intensity falls to $1/e$. The last column is the difference between the two. Points 11 to 20 share the optical layout of point 10, so their predicted and measured sizes are identical and have been omitted.

Point	$f_{\#}$ [-]	M_{agexp} [-]	Δs_{th} [px]	Δs_{exp} [px]	Δs dev. [px]
1	16.00	1.002	3.72	20.30	+16.58
2	16.00	0.688	3.15	13.05	+9.90
3	16.00	1.045	3.90	21.21	+17.31
4	16.00	0.671	3.12	12.74	+9.62
5	11.31	1.043	2.74	20.78	+18.04
6	11.31	0.718	2.27	14.12	+11.85
7	11.31	1.054	3.02	25.32	+22.30
8	4.00	0.572	0.75	12.35	+11.60
9	16.00	0.640	3.06	12.65	+9.59
10	8.00	0.746	1.65	12.94	+11.29

In this section, the results, errors and associated uncertainties of the angled glass experiments have been presented. The two plots constructed with the data collected permit answering one of this thesis' research questions and infer the mechanism that explains the advantage in performance of SBOS in comparison to BOS. Nevertheless, there was one prediction from theory that could not be verified in this experimental campaign: the average speckle size given by Rayleigh Diffraction. The speckles obtained were considerably larger and did not change with the lens f-stop number, which led to the belief that the laser ray was becoming diffraction limited in the ground glass diffuser instead of the lens pupil. Another experimental campaign was envisioned to try and obtain a finer speckle pattern before the implementation of SBOS to a wind-tunnel application.

6.2. Speckle Pattern Investigation Results

Faced with the difficulties in creating a satisfactory speckle pattern in the first experimental campaign, the second experiment aimed to investigate ways to generate a finer average speckle size, try to produce a Rayleigh distribution at the lens pupil and compare the implications of generating a speckle through direct diffusion or by projecting the laser into a surface. Table 6.7 presents the data obtained with the experimental setups described in Section 5.2 with the conditions displayed in Tables 5.2 for direct speckles and 5.3 for projected speckles. This section discusses the computed average speckle sizes obtained at each point, the deviation from the Rayleigh limit of Equation 4.9 and a quantification of the experiment's associated uncertainties.

Table 6.7 and Figure 6.2 show the average speckle size obtained experimentally for each point. They present the experimental speckle sizes obtained using DaVis' function together with the ones computed through Hamarová et al.'s autocorrelation method [18]. A comparison between the two methods yielded some differences in magnitudes, with DaVis ranging from 1.19 to 2.41 px, while the autocorrelation spans from 1.62 to 4.59 px. Despite this difference, both methods reproduce the same trends quite reliably as shown by Figure 6.2. Since it is known how the estimator of Hamarová et al. works, contrary to the DaVis estimator, and both reproduce the same behavior, from now on the discussion will focus only on the results obtained with the autocorrelation method.

Regarding the Rayleigh limit reproduction, it is possible to see that the results are much more similar to the predicted values in order of magnitude if compared to the ones presented previously in Table 6.6. The deviations now span from 0.06 px to a maximum of 2.88 px, instead of 9.59 to 22.30 px from before. Nevertheless, the relative error is still large overall, with the worst results at the points with

Table 6.7: Speckle sizes measured during the experimental campaign, using the optical configurations of Table 5.2. M_{agexp} is the magnification measured with the calibration plate, Δs_{th} is the size predicted by the Rayleigh limit and Δs_{exp} is the estimated experimental size, reported both by the DaVis function and by the method of Hamarová et al. [18]. The deviation columns give the difference of each measurement from the prediction. The column u_{tot} is the combined standard uncertainty of Δs_{exp} . Points 1 to 13 vary the aperture and the ground glass distance, whereas points 14 to 28 hold both fixed and sweep the magnification. Points 29 to 31 were obtained with projected speckles, using the parameters of Table 5.3.

Point	$f_{\#}$ [–]	d_{GG} [mm]	M_{agexp} [–]	Δs_{th} [px]	Hamarová			DaVis	
					Δs_{exp} [px]	Δs dev. [px]	u_{tot} [%]	Δs_{exp} [px]	Δs dev. [px]
1	4.00	740	0.964	0.92	1.74	+0.82	6.35	1.20	+0.28
2	11.31	740	0.964	2.59	2.73	+0.14	1.50	1.48	–1.11
3	22.63	740	0.964	5.19	3.83	–1.36	11.45	2.00	–3.19
4	4.00	740	0.296	0.61	1.62	+1.02	2.62	1.19	+0.59
5	11.31	740	0.296	1.71	2.71	+1.00	7.11	1.56	–0.15
6	22.63	740	0.296	3.42	3.81	+0.39	4.87	2.04	–1.39
7	11.31	450	0.296	1.71	2.29	+0.58	6.60	1.35	–0.36
8	11.31	280	0.296	1.71	2.45	+0.74	2.87	1.40	–0.31
9	11.31	190	0.296	1.71	2.91	+1.20	2.79	1.59	–0.12
10	11.31	100	0.296	1.71	4.59	+2.88	3.12	failed	–
11	4.00	300	0.877	0.88	2.42	+1.54	1.50	1.45	+0.57
12	11.31	300	0.877	2.49	2.98	+0.49	1.50	1.67	–0.82
13	22.63	300	0.877	4.97	4.15	–0.83	1.55	2.41	–2.56
14	22.63	300	1.000	5.37	3.05	–2.33	1.52	1.81	–3.57
15	22.63	300	0.909	5.13	3.14	–1.99	1.50	1.76	–3.38
16	22.63	300	0.800	4.84	3.26	–1.58	1.50	1.86	–2.98
17	22.63	300	0.667	4.48	2.66	–1.81	6.43	1.60	–2.88
18	22.63	300	0.571	4.22	2.80	–1.42	4.78	1.65	–2.57
19	22.63	300	0.500	4.03	2.39	–1.64	2.80	1.47	–2.56
20	22.63	300	0.400	3.76	3.60	–0.16	2.39	1.68	–2.08
21	22.63	300	0.333	3.58	2.16	–1.42	4.13	1.42	–2.16
22	22.63	300	0.286	3.46	2.38	–1.07	3.39	1.48	–1.97
23	22.63	300	0.222	3.28	2.53	–0.76	4.66	1.59	–1.70
24	22.63	300	0.167	3.13	2.88	–0.25	3.33	1.74	–1.39
25	22.63	300	0.118	3.00	3.06	+0.06	3.42	1.91	–1.09
26	22.63	300	0.074	2.89	2.72	–0.17	3.05	1.65	–1.23
27	22.63	300	0.054	2.83	2.76	–0.07	1.82	1.71	–1.12
28	22.63	300	0.035	2.78	3.09	+0.31	5.47	1.87	–0.91
29	4.00	300	0.457	0.67	1.59	+0.92	1.54	1.22	+0.55
30	11.31	300	0.457	1.90	2.21	+0.31	1.80	1.44	–0.46
31	22.63	300	0.457	3.81	3.61	–0.20	2.08	2.09	–1.72

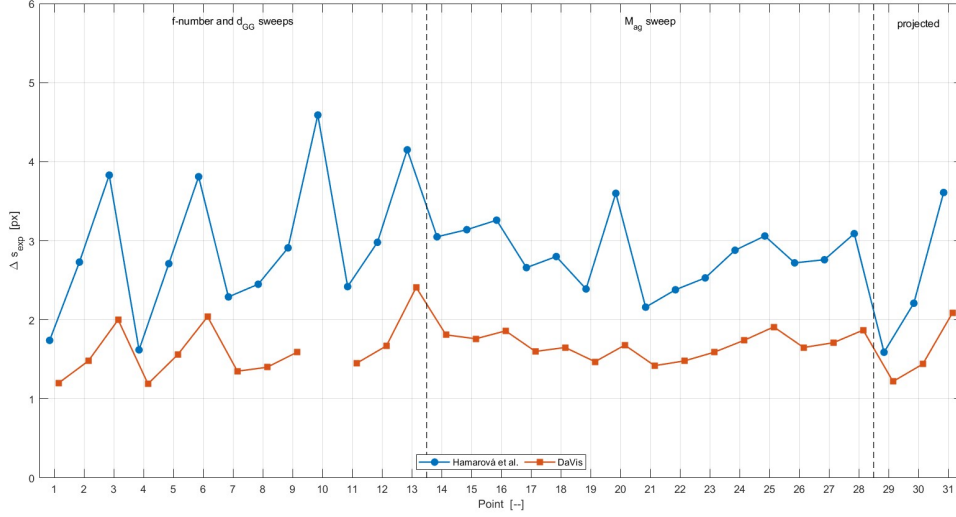


Figure 6.2: Average speckle size of every point of Table 6.7, computed with the autocorrelation method of Hamarová et al. [18] and with the DaVis function. The dashed lines separate the $f_{\#}$ and d_{GG} sweeps, the magnification sweep and the projected speckle points. DaVis returned no value for point 10

subpixel speckles and close to 5 px. The error also vastly surpasses the uncertainty associated with magnification, which stayed within 5% for a 10% deviation in M_{ag} , and post-processing, which had a maximum of 11.45%. Regardless, the data obtained provides a better fit to the Rayleigh diffraction prediction, showing that the equation can be used as a rough preliminary estimator of the speckle size. The results also allow for the identification of useful trends and comparisons in speckle production, as will be shown.

Points 29 to 31 of Table 6.7 were obtained with projected speckles. Compared to direct speckles, the projected speckles show a similar deviation to theory and an overall smaller speckle size, which is hinted by Figure 6.3. The results are too similar to the direct speckle approach to discriminate which method to use based on speckle size. Therefore, the trade-off on how to create the speckle pattern depends more on practical considerations than on its speckle characteristics. Projected speckles might be preferable in applications where only one optical access is available; where powerful lasers are being used that could damage the camera or when the double-pass method of Meier & Rösger [32] needs to be used for increased sensitivity. Direct speckles, on the other hand, provide more intensity, which allows for lower exposures, and might be preferable when a see-through optical access is available.

To better judge the deviation of the experiment, an uncertainty budget was constructed to compute the total uncertainties. They are displayed for each experimental point in Table 6.7. This total uncertainty corresponds to the ones obtained for the implementation of Hamarová et al.'s method and is reported in Table 6.8. In their paper, the researchers reported an uncertainty of 1.5% associated with their estimator. As a consequence, the budget is dominated by the Gaussian filter implemented. The Gaussian standard deviation was taken as a parameter and varied such that the tail of the autocorrelation function result after 60 px was minimized. The half-width of the method is, therefore, the achieved standard deviation and its individual uncertainty, assuming a rectangular distribution, is $u = a/\sqrt{3}$. The contribution is u times the sensitivity coefficient. In the last row, the combined figure of the post-processing uncertainty can be found.

Now that the error and uncertainties of the results have been discussed, the data can be plotted to better understand its behavior, such as f-stop dependency. The various $f_{\#}$ sweeps made during the experimental campaign allow the evaluation of whether the interference waves became diffraction limited at the pupil. Figure 6.3 shows the trend of average speckle size with f-stop number for four different configurations. All lines follow the characteristic increasing trend of a developed subjective speckle, which follows Equation 4.9, and signals that the interference front is becoming diffraction limited at the

Table 6.8: Uncertainty budget for the average speckle size obtained by the autocorrelation method. The flattening filter width is not measured with an instrument; it is selected by driving the tail of the autocorrelation to zero, so its half-width a is the range of widths for which that tail remains within its noise floor, taken as $1/\sqrt{N_s}$ for a frame holding N_s speckles. As in Table 6.4, the width is treated as a rectangular distribution so that $u = a/\sqrt{3}$ [26], and the contribution c_σ is the sensitivity $\partial\Delta s_{exp}/\partial\sigma$, obtained by re-evaluating the speckle size one pixel either side of the selected width, multiplied by u . The estimator contribution is the spread between the two-dimensional, the average one-dimensional and the long vector estimators reported by Hamarová et al. [18]. Both terms vary per point, so their range over the 31 points is given here and the totals are listed per point in Table 6.7.

Source	Instrument	a	u	Contribution c_i [%]
Filter width σ	Tail criterion	2.0–67.0 px	1.2–38.7 px	0.06–11.36
Estimator choice	Literature	–	1.5 %	1.50
Combined, $u_{\Delta s} = \sqrt{c_\sigma^2 + c_{est}^2}$				1.50–11.45

lens pupil [15]. The red and yellow lines both share the same d_{GG} , which makes them viable for direct comparison. Curiously, it appears that the increased magnification of the red line caused practically no effect in the average speckle size, displaying almost the same values as the yellow line. The same insensitivity to magnification was also experienced through points 14–28 shown in Figure 6.6. Looking at the purple line compared to the latter two, it appears that the different d_{GG} introduced some difference in the speckle size, with the purple exceeding the red by 0.68, 0.25 and 0.32 px for $f_\# = 4$, 11.31 and 22.63, respectively, although the difference is only subpixel. In green, the projected speckle trend can also be analyzed, and it demonstrates that projected speckles are generally smaller, presenting smaller speckles even than the yellow line, which has a lower magnification than green. The figure shows some promising trends as there is a clear dependence on $f_\#$ and it indicates a tendency of projected speckles to be smaller than direct ones. However, it also shows an insensitivity to the optical magnification, contradicting Equation 4.9, and the effect of d_{GG} which will be further discussed next.

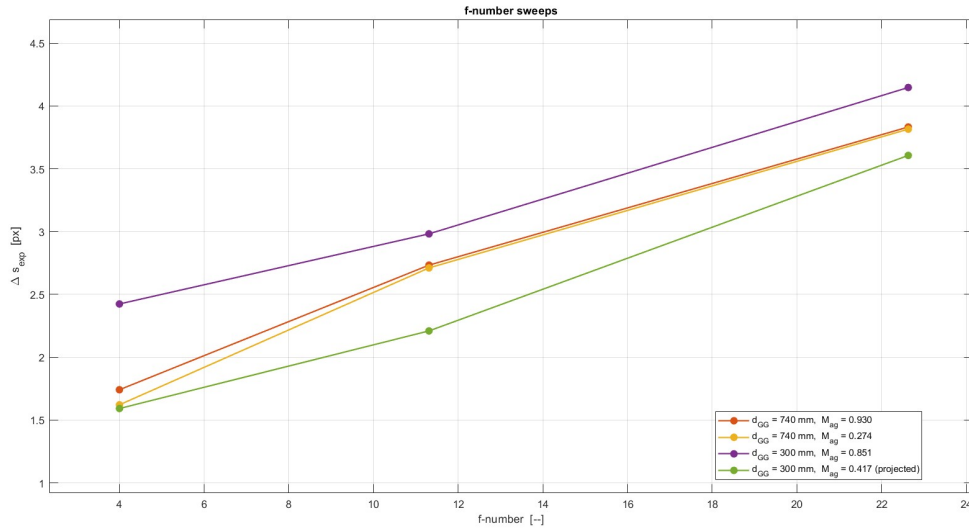


Figure 6.3: Average speckle size versus f-stop number for the four aperture sweeps of the speckle investigation campaign. The red and yellow lines correspond to points 1–3 and 4–6, which share $d_{GG} = 740$ mm at two different magnifications. The purple line corresponds to points 11–13, acquired at $d_{GG} = 300$ mm, and the green line to points 29–31, obtained with projected speckles at the same d_{GG} .

Since the beam was expanded through a convergent lens, the distance between that lens and the ground glass diffuser is a proxy for the beam diameter, therefore, it is possible to verify the influence of the beam diameter in the speckle generation by plotting the points where only d_{GG} was varied. Figure 6.4 shows that the average speckle size remains quite consistent for high d_{GG} , as it varies from 2.91 px at 190 mm, dips to 2.29 px at 450 mm and goes to 2.71 px at 740 mm. This variation is consistent with the findings of Figure 6.3. It demonstrates that d_{GG} can increase the speckle size but for large

distances the spread is practically subpixel. However, the figure also shows a spike at low d_{GG} when the beam diameter is small, reaching 4.59 px at 100 mm with beam ratio of 0.25. For the DaVis function, the envelope was so small and the speckle so big that it failed to calculate it. The effect can be better visualized through Figure 6.5 which displays side-by-side the captured speckle patterns for points 5, 8 and 10 respectively. This data supports the hypothesis that the larger speckle size in the angled glass experiment was caused by the small diameter of the laser when it hit the 1.27 cm ground glass diffuser. The fact that the speckle size changed less for larger distances is also backed up by the literature, which does not mention the effect of the laser diameter in the speckle generation. In addition, it means that, when designing the laser optical setup, an experimentalist must only worry about expanding the laser beam to fill the desired field of view.

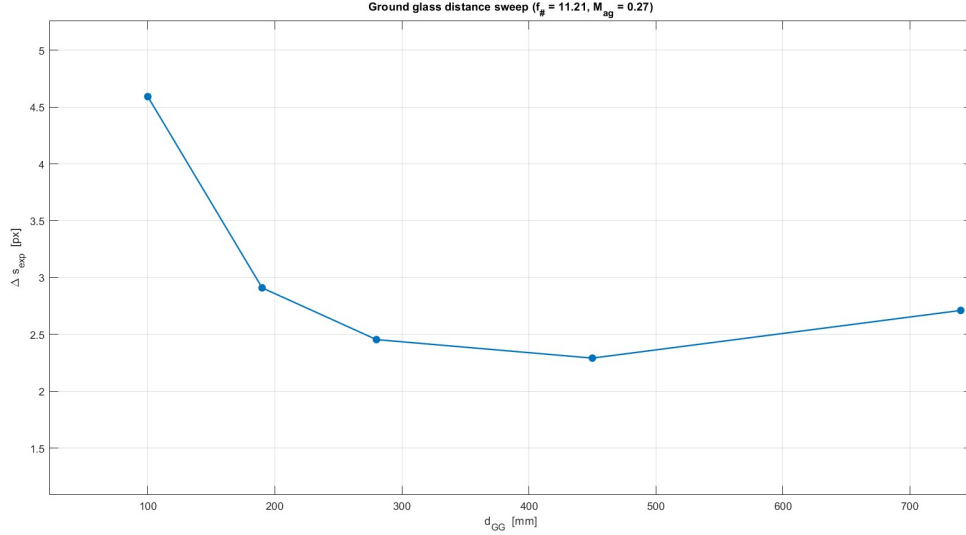


Figure 6.4: Average speckle size versus the distance between the beam expander lens and the ground glass diffuser, for points 5 and 7-10 of Table 6.7, all acquired at $f_{\#} = 11.31$ and $M_{agexp} = 0.296$

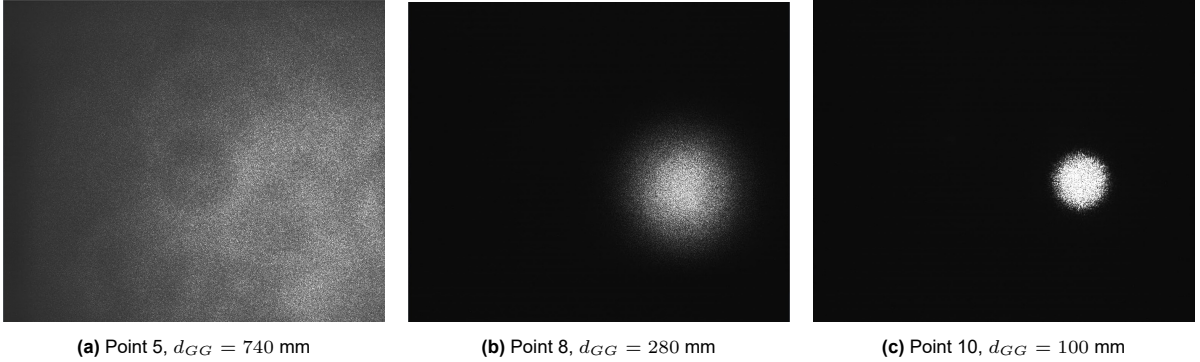


Figure 6.5: Raw speckle patterns captured at $f_{\#} = 11.31$ and $M_{agexp} = 0.296$ for three distances between the beam expander lens and the ground glass diffuser. As d_{GG} decreases, the illuminated area on the diffuser shrinks and the speckle pattern contracts from filling the sensor to a small bright spot

To better assess the effect of optical magnification in the speckle generation process, a sweep of the focal distance was conducted through points 14-28 and is shown in Figure 6.6. It is worth mentioning that the magnification value of these points is less accurate than the others, as the focal distance was obtained through the distance marks of the camera lens. Nevertheless, the figure shows, for a different d_{GG} , the same speckle size insensitivity to optical magnification that was identified in Figure 6.3. That effectively contradicts the Rayleigh diffraction limit equation and reduces the number of control variables an experimentalist would have when designing the experimental setup.

The speckle pattern investigation campaign allowed for the study of the speckle generation process. It

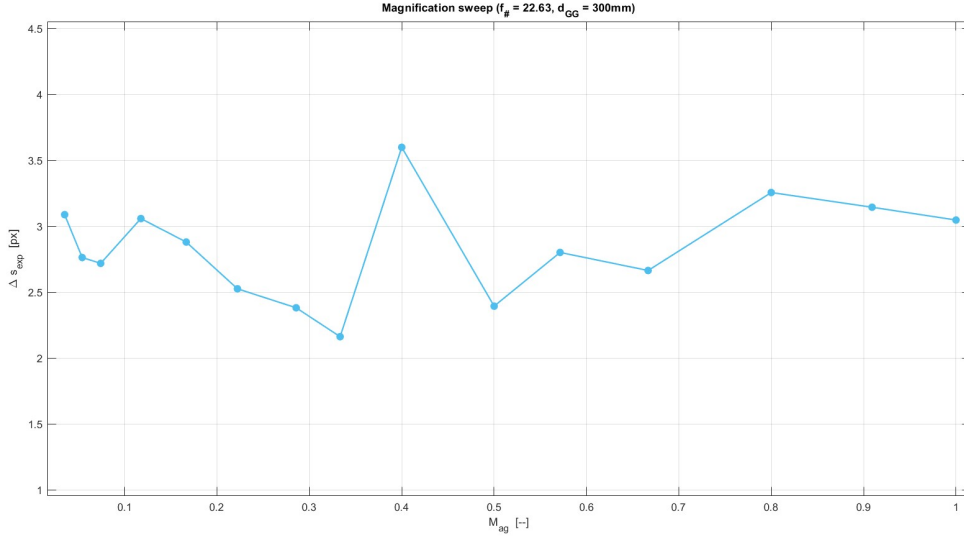


Figure 6.6: Average speckle size versus optical magnification for points 14-28 of Table 6.7, acquired at a fixed $f_{\#} = 22.63$ and $d_{GG} = 300$ mm. The magnification of these points was obtained from the distance marks of the camera lens rather than with the calibration plate

successfully reproduced finer speckles than the angled glass experiment, showed the dependence of the speckle size on the f-stop number, investigated the effect of a small incidence beam in the diffuser and compared direct with projected speckle. However, the data showed unexpected insensitivity to magnification variations and still deviated significantly from the speckle predicted through the Rayleigh limit. It is worth reiterating that the speckle generation process is a complex intrinsically random phenomenon and to treat it with much more detail would be out of the scope of this thesis. For now, the data at hand make it possible to define a laser speckle generator capable of producing speckles within the 2-5 pixel range. Therefore, the findings of this campaign are enough to proceed to the correct application of Laser Speckled BOS to image a real schlieren object, in this case a compressible jet.

6.3. Compressible Jet Experiment Results

An underexpanded compressible jet was chosen as the object of study for the implementation of SBOS to a real schlieren object. The goal of this experimental was the imaging of the jet at various nozzle pressure ratios (NPR), post-process the raw displacement data into a quantitative density field, analyze the jet structure and study the influence of an increase in nozzle pressure. The setup of this experiment was explained in section 5.3. Figure 6.7 collects the results of the cross correlation for the 6 different gauge pressures set at the wall air tap.

Comparing the different images displayed by Figure 6.7, it is possible to assess the effect of different NPR to the underexpanded jet structure. As explained in section 2.2, an increase in NPR leads to an increase in shock cell length and a fundamental change in the jet structure at certain pressure ranges. Comparing the result for $p_g = 3$ bar to $p_g = 8$ bar, the shock cell length does indeed increase, together with the number of cells and a bigger bulge mid-cell. However, there is very little change in the shock cell length for gauge pressures 6 through 8 bar. Moreover, from Figure 6.7b onward the NPR is within 4 to 7 so the formation of a Mach disk together with a "barrel" shaped shock cell would be expected, see Figure 2.5b. Nevertheless, all images seem to display an "X" shape, related to a nozzle pressure ratio of 2 to 4, Figure 2.5a. Together, these deviations from theory point that the real pressure ratios at the nozzle are not what they would be if the total pressure were maintained from the wall air tap gauge, through the hose and out of the air gun. Therefore, there must be heavy pressure losses present throughout the line that need to be accounted for and new estimates of the nozzle pressure ratios must be computed.

The strategy envisioned to better estimate the nozzle pressure ratio builds on top of the definition of the

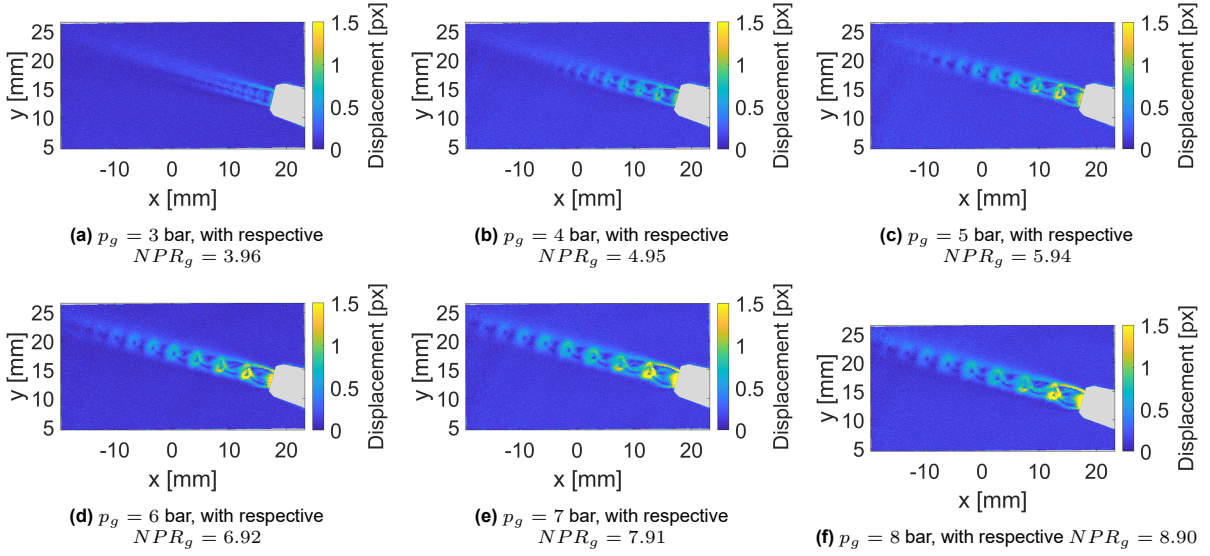


Figure 6.7: SBOS results for a compressible jet. Displacement magnitude fields obtained through cross correlation for air tap gauge pressures from 3 bar to 8 bar with a step of 1 bar. All fields share the same colour scale and masked vectors are shown in grey

Prandtl-Meyer angle. As explained in section 2.1, ν is the turning angle needed for a flow at a certain Mach number M_1 to expand to a Mach number M_2 . In the case studied, the flow is choked at the nozzle exit, so the Mach number at the exit is assumed to be equal to one $M_e = 1$. At the lip, the flow expands isentropically until it reaches a certain M_j at ambient pressure. Therefore, the angle of the flow at the shear layer is equal to $\varphi = \nu(M_j)$ relative to the jet axis. So, to better estimate the NPR , the images were then treated to make the jet axis horizontal, then the shear layer angle at the tip was measured, M_j was calculated through the Prandtl-Meyer function and a value for p_e was obtained using isentropic expansion. The values can be found in Table 6.9 together with the uncertainty corresponding to $+1^\circ$ in ν .

Table 6.9: Estimated jet conditions for each Prandtl-Meyer angle ν . M_j is the jet Mach number, NPR the nozzle pressure ratio and p_0 the absolute total pressure. The uncertainties u_{M_j} , u_{NPR} and u_{p_0} are given relative to their respective values. The data is organized in letters from "a)" to "f)", corresponding to the order of the images in Figure 6.7

Data	ν [$^\circ$]	M_j [-]	u_{M_j} [%]	NPR [-]	u_{NPR} [%]	p_0 [bar]	u_{p_0} [%]
a)	1	1.082	4.71	2.088	6.51	2.115	6.52
b)	3	1.177	3.74	2.353	5.48	2.384	5.49
c)	5	1.257	3.10	2.613	5.17	2.647	5.14
d)	8	1.366	2.64	3.032	5.01	3.072	5.01
e)	10	1.435	2.44	3.344	5.02	3.387	5.02
f)	13	1.537	2.21	3.875	5.08	3.925	5.10

The NPR estimates displayed in Table 6.9 better fit the theory but also means that the pressure losses in the line were extremely high. The nozzle pressure ratios range from 2 to 4, exactly the range where an "X" shaped shock cell would be expected. Moreover, the pressures of datasets e and f change very little between them, while there is a big difference between a and f, which is coherent with the similar pictures obtained. A question arises, however, when the estimate of total pressure is compared with the one at the air tap gauge, as loss of over 40% can't be explained with friction losses alone. These other sources of loss were probably choke points in the line, at quick disconnect joints or at a non-fully pressed air gun trigger. Another way to verify these estimates is by computing the density gradient field, measuring the shock cell length and relating them to the NPR through the relationships of section 2.2.

A density gradient field can be obtained assuming a planar field, through equation 2.22. This result

corresponds to the projection of the axysymmetric body. In reality, the thickness of the object W and density gradient varies with the radius, so this solution will underestimate the density gradients at the borders of the jet. Nevertheless, it is useful to qualitatively study the shock cell structure, as the periodicity of the cell and the gradient signs are maintained in the projection. Figure 6.8 shows the gradients for dataset f in the two directions s and n together with the magnitude of the vector.

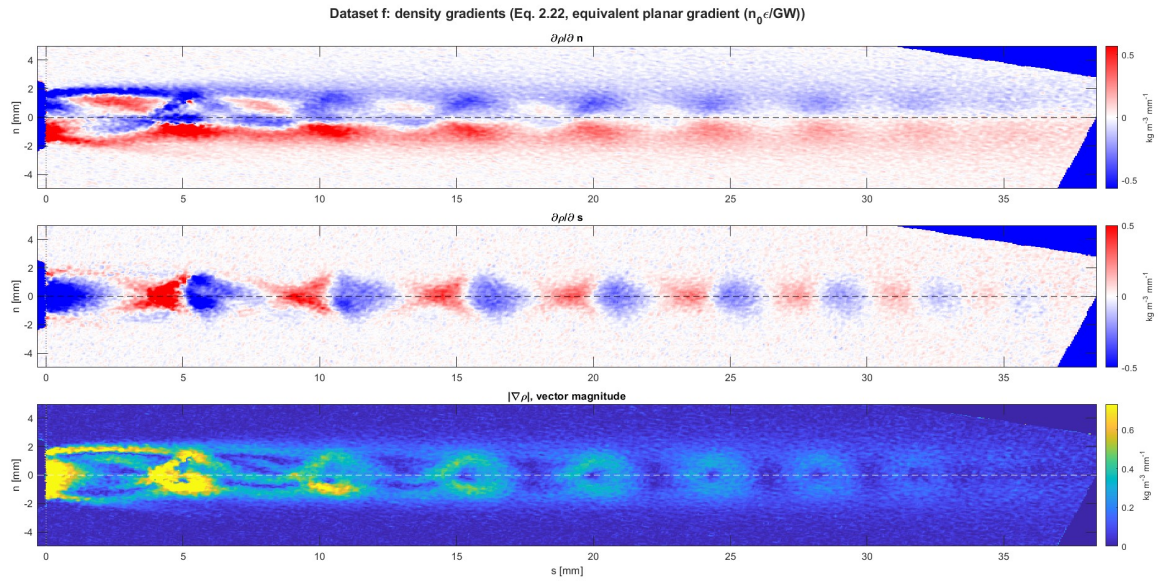


Figure 6.8: Density gradient fields of dataset f obtained assuming a planar field. The top plot displays the density gradient in the n direction, the middle plot the density gradient in the s direction and the bottom plot the magnitude of the density gradient vector. The s axis is aligned with the jet axis with its origin at the nozzle exit, while n is normal to it

Following the middle plot of Figure 6.8, correspondent to the density gradient in the s direction, the jet structures can be easily identified. Spanning from the lip to the axis at the nozzle exit, an expansion fan causes the density to decrease severely, indicated by the blue region in the core of the jet. A little above the lip, an oblique shock is marked in red going from the lip to the axis downstream, where it reflects in direction to the shear layer, forming an "X" shape. This intercepting shock produces a region of increasing density, marked in red. After the density peaks, a new round of expansion waves causes the density to decrease, thus starting a new shock cell. The periodicity of shock cells are also evident if the centerline density gradient is plotted, Figure 6.9 displays such graph for the 6 datasets.

The 6 plots in Figure 6.9 permit the identification of important marks in the shock cell structure. The periodicity and dissipation effect is clear, each cell is marked by an expansion valley and a compression peak and its amplitude decreases until atmospheric conditions are reached. The density maxima and minima are also distinguishable as the points where the gradient is zero, the distance between the first

6.4. Aerospike Experiment Results

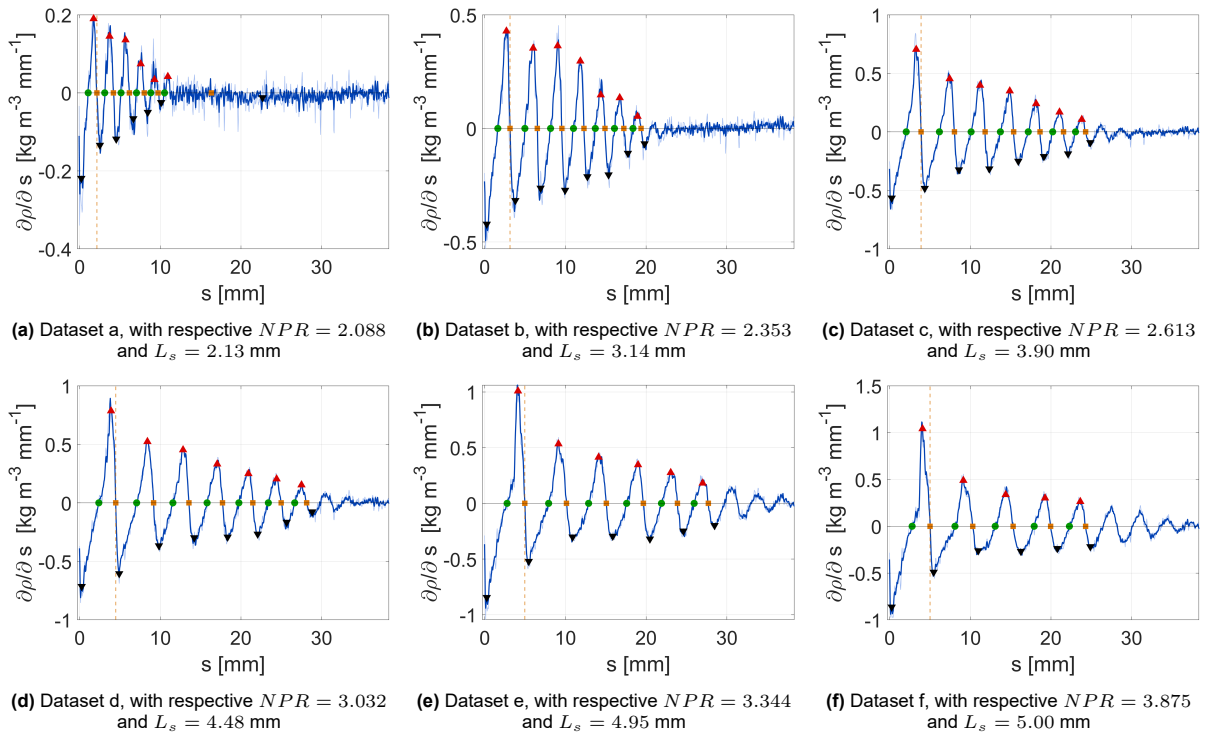


Figure 6.9: Centerline density gradient in the s direction for datasets a to f, obtained assuming a planar field. The light blue line is the gradient at the axis row $n = 0$ and the dark blue line its mean over $|n| \leq 0.25$ mm. Red upward triangles mark the compression peaks, black downward triangles the expansion valleys, green circles the density minima and orange squares the density maxima detected along the centerline. The dashed orange line marks the end of the first shock cell, L_s

7

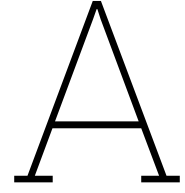
Conclusion

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Derivation of the Circle of Confusion

The Circle of Confusion estimates through classical optical theory the amount of blur present in the image of an out of focus object, as explained by Greenleaf [16]. As shown in Figure A.1, an object O at a distance d_f of a converging lens, with focal length f and aperture D , produces an image I at a distance Z of this lens, as per the ideal lens equation. If an object O' finds itself behind of O by a defocus distance l , its image will form at Z' on the opposite side of the converging lens. However, a point in object O' will appear smeared into a circle in the plane of image I , as it is out of focus. To calculate the size of this circle, classic optics and geometry can be used.

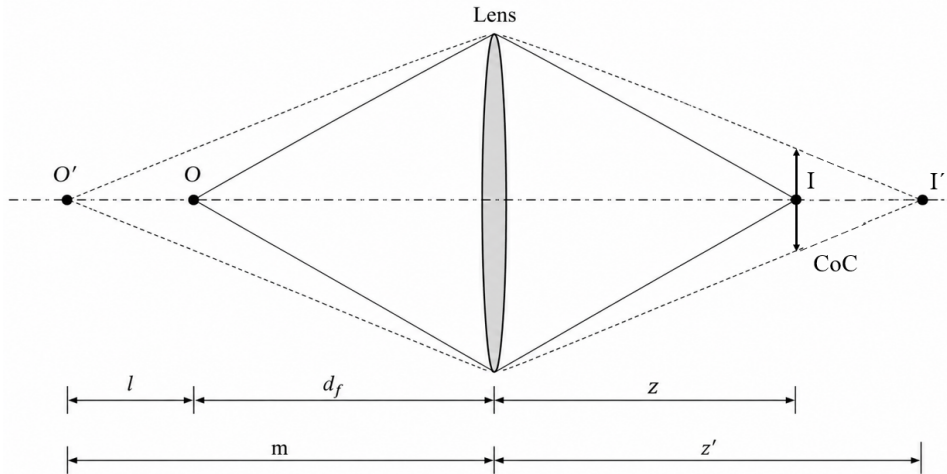


Figure A.1: Geometry of an out of focus object used to derive the CoC

Through the ideal lens equation, Z and Z' can be written in the following form:

$$Z = \frac{f d_f}{d_f - f} \qquad Z' = \frac{f m}{m - f} \qquad (\text{A.1})$$

Using triangle similarity, the relationship below can be established:

$$\frac{\text{CoC}}{Z' - Z} = \frac{D}{Z'} \qquad (\text{A.2})$$

Substituting Equation A.1 in Equation A.2 and doing the necessary simplification, an explicit equation for CoC is obtained:

$$CoC = \frac{f^2 l}{f_{\#} m (d_f - f)} \quad (A.3)$$

Where $f_{\#} = \frac{f}{D}$ is the lens f-stop number.

In Chapter 4, the sign of l was considered to change depending on its position relative to the in-focus plane. This way, if an object finds itself behind the in-focus plane, Equation A.3 will result in a negative CoC , while if the object is in front, it will be positive. Since the circle of confusion is a length, its absolute value can be considered in these cases.

Equation A.3 can also be modified by substituting it with Equation A.4, which define the magnification M_{ag} and sensitivity S

$$M_{ag} = \frac{f}{d_f - f} \quad S = l M_{ag} \quad (A.4)$$

Resulting in:

$$CoC = \frac{f l M_{ag}}{f_{\#} m} \quad CoC = \frac{f S}{f_{\#} m} \quad (A.5)$$

The CoC is also in the core of another important parameter in optics called depth of field (DoF) [16]. The depth of field is the region around the in-focus plane where the CoC is below a threshold deemed acceptable. This way, the depth of field can be interpreted as the thickness of the in-focus plane, where the image of all the objects inside this region is considered sharp enough. By definition, DoF can be calculated by subtracting the object distances m_+ and m_- that produces the same CoC when in front of the in-focus plane and behind it, respectively.

B

Optics of a Blurred Object

When an imaged object is blurred, its features do not converge to a single distinct point but are instead smeared around an area called circle of confusion. Figure B.1 shows an object of height H' and how light travels from the highest point of the object to interact with the convergent lens in the sketch. The image also shows an object in focus with height H and its image height h . Notice how the light rays from the in-focus object make a triangle with its image, it's this geometry that allows for the derivation of relationship between distances d_f , f and Z called thin lens equation, described in Equation 4.2. For an out-of-focus object, the relationship between object distance m , image distance Z and f is no longer described through the thin lens equation as the geometry of the system has changed. Nevertheless, It is possible to use this new geometry to define new concepts, such as the blurred object magnification $M_{ag_{obj}}$, which corresponds to the ratio between the size of the out-of-focus object and the image of the in-focus object that shares the light ray that crosses the middle of the lens with the out-of-focus object.

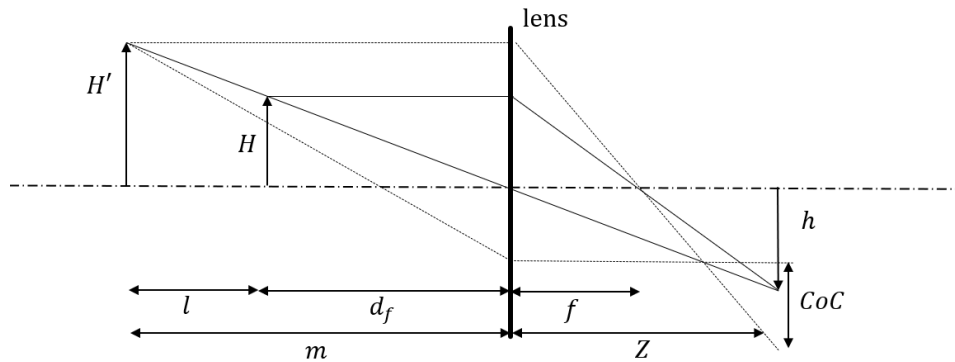


Figure B.1: Geometry of an out of focus object

$$M_{ag_{obj}} = \frac{h}{H'} = \frac{Z}{m} \quad (\text{B.1})$$

This magnification can also be written in terms of the focal length and the in-focus magnification:

$$M_{ag_{obj}} = \frac{f(1 + M_{ag})}{m} \quad (\text{B.2})$$

Equivalently to how M_{ag} can be used to transfer a length from the image plane to the object plane and vice-versa by dividing or multiplying it by M_{ag} , respectively. The same can be done with $M_{ag_{obj}}$. This way, it is possible to transfer the CoC from the image plane to the blurred object plane. In this same way, $L_{FoV_{obj}}$ was derived in Equation 4.8.

$$CoC_{obj} = \frac{CoC}{M_{ag_{obj}}} \quad (B.3)$$

By combining Equations A.5, B.2 and B.3, it is possible to obtain:

$$CoC_{obj} = \frac{lM_{ag}}{f_{\#}(1 + M_{ag})} \quad (B.4)$$

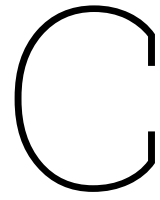
and rearranging Equations B.3, B.1 and 4.1:

$$CoC_{obj} = \frac{CoC}{M_{ag}} \frac{m}{d_f} = \frac{CoC}{M_{ag}} \left(1 - \frac{l}{m + l} \right) \quad (B.5)$$

Given the spatial resolution δ_t derived by Gojani & Obayashi, described by equation 10 on their article [14].

$$\delta_t = \frac{lM_{ag}}{f_{\#}(1 + M_{ag})} + \frac{\Delta_t}{M_{ag}} \left(1 - \frac{l}{m + l} \right) \quad (B.6)$$

If Δ_t corresponds to the CoC as it is mentioned in Gojani's paper, his blurred circle becomes $\delta_t = 2 \cdot CoC_{obj}$ through Equations B.4 and B.5. If the geometry of Gojani's cone of light is also investigated a bit further, it is possible to notice that it resembles the geometry of Figure A.1 but off the lens axis and mirrored around the lens' plane. This observation leads to the interpretation of Gojani's result as a calculation of the Circle of Confusion in the object plane of two points in the image plane that form the Circle of Confusion, or the object Circle of Confusion of the image Circle of Confusion, which is...a rather confusing result. Nevertheless, It is thus proven that Gojani's formulation predicts a blur circle twice the size of the one established experimentally by Schwarz and Braukmann [14, 44].



Extended Surface Maps for SBOS

This appendix collects the surface maps that were computed by the design space exploration of Chapter 4 but were not reproduced in the main text. They are provided for completeness and to support the validation of the design strategy during the first experimental campaign.